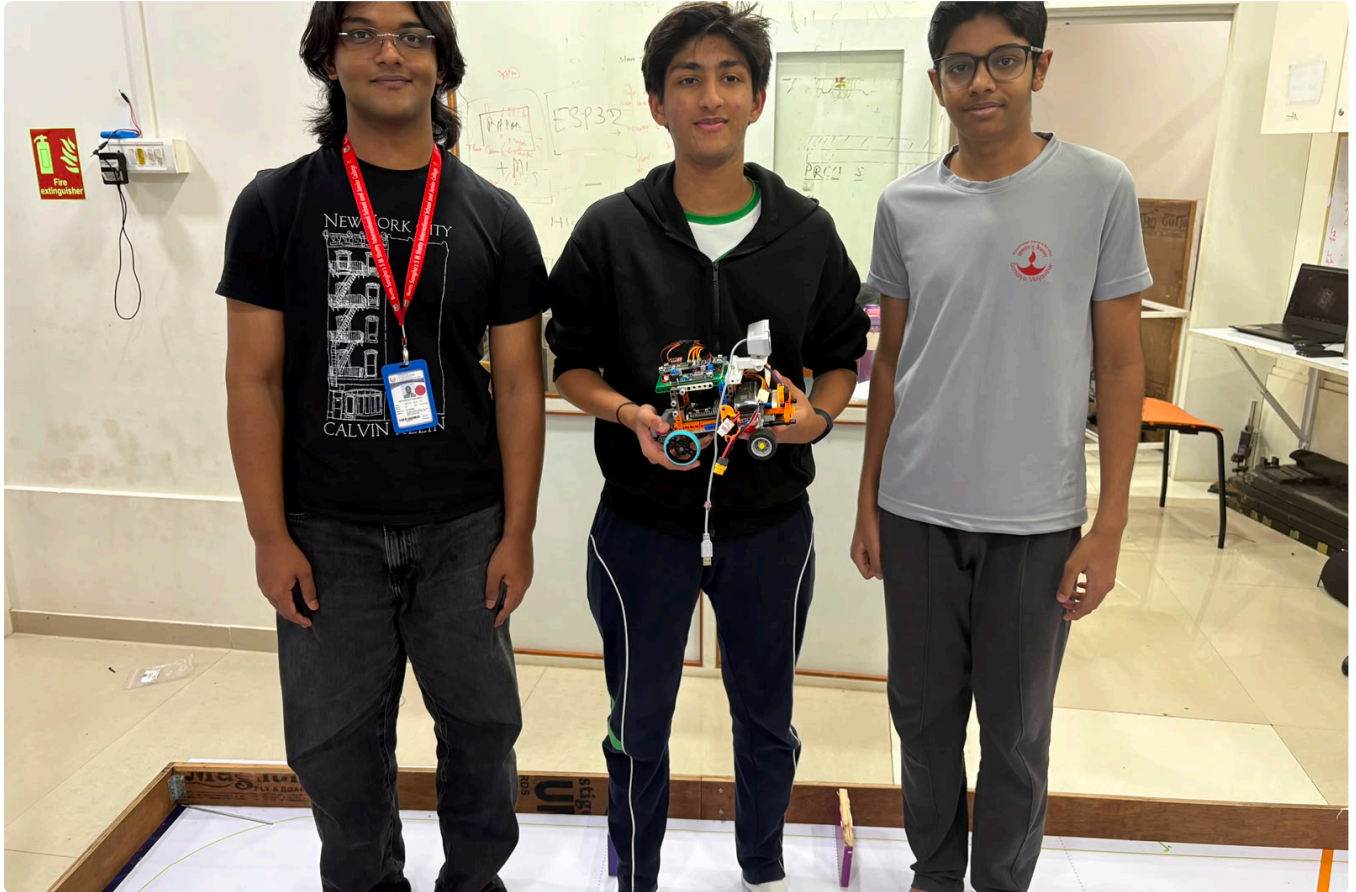


WRO FUTURE ENGINEERS 2026 · APAC

# Engineering Journal

APAC 2026 edition (rev 4) · Team TED Drive



|             |                                                                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Team        | Ethan Fernandes (software) · Tejas Sirikonda (mechanical) · Diaan (electronics)                                                                   |
| Coach       | Amey Chavan (OMOTEC)                                                                                                                              |
| Competition | WRO Future Engineers 2026, APAC, Hyderabad, September 2026                                                                                        |
| Repository  | <a href="https://github.com/teddriveomo/wro-2026-future-engineers">github.com/teddriveomo/wro-2026-future-engineers</a> (release 0.5.0 and later) |
| Prepared    | 23 September 2026, against WRO 2026 General Rules, Appendix C                                                                                     |

Part I describes the vehicle rebuilt for APAC and is generated from the repository's own documents, so the journal and the repository say the same thing. Part II is the India Nationals journal (rev 3, 24 August 2026), kept unchanged as the record of the first vehicle.

# 01 Snapshot

The APAC vehicle in one page; every figure is from Part I and the repository.

|          |                                                                                                                                                                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vehicle  | Rebuilt in September 2026: a new 84-step LEGO Technic frame (332 g bare), a steering column that turns the whole front axle 60° each way, the camera on a 24.8 cm tower pointing 12° down, six WS2812 LEDs on the front frame. The horizontal side rollers of the rebuild were removed on 22 September.                      |
| Mass     | 862 g race-ready with the Raspberry Pi 5 (measured 16 September, before the rollers came off).                                                                                                                                                                                                                               |
| Drive    | MEX 6 V motor through a TB6612FNG driver to the rear wheels: 422 rpm free-running at the wheel and 0.89 m/s average over 3 m from a standing start (measured 21 September).                                                                                                                                                  |
| Power    | 3S 2200 mAh pack: 1.34 A idle, 1.93-2.02 A driving, 2.77 A worst case (each branch measured, 21 September).                                                                                                                                                                                                                  |
| Compute  | Raspberry Pi 5 (camera, detection, decisions) and ESP32 (drive motor, servo, three TF-Luna, LEDs), linked by USB serial.                                                                                                                                                                                                     |
| Software | <code>main.py</code> dated 18 September: Lab-colour pillar detection with tape-line corner context, a state machine that recentres between the walls after every pillar and corner, two pillars per straight, and a CSV log of every run. Parking at the end: <code>parallel_parking.py</code> , the team's choice for APAC. |
| Evidence | Bench measurements of drive, power and camera geometry (§02, §03); detection shown on the mat (§04); code stored byte-for-byte with MD5 checksums (§06).                                                                                                                                                                     |

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| II | India Nationals journal, rev 3 (the first vehicle)                                             | C1-C5      |

## 02 Mobility

*How the car is built and moves (Appendix C, criterion 1).*

**Status 2026-09-22.** After the WRO India National Championship (26–28 Aug 2026) the vehicle was rebuilt for APAC 2026. This page records what changed and why. Every figure is marked by its basis: *measured* on the vehicle, *team* figure not yet measured, *derived* by calculation from named inputs, or *vendor* specification. Documents 1–5 remain the record of the Nationals configuration.

### 1. Specification

|                         | Nationals (Aug 2026)                                                                                                                             | APAC 2026 (Sep 2026)                                                                                                                                                        | Basis                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Length × width × height | length 23.0 cm                                                                                                                                   | 19.1 × 14.3 × 27.2 cm (±0.5 cm)                                                                                                                                             | measured             |
| Mass                    | 785 g                                                                                                                                            | 862 g race-ready, Raspberry Pi 5 included (2026-09-16); not re-weighed after the side rollers were removed (§7)                                                             | measured             |
| Axle loads              | —                                                                                                                                                | front 201 g, rear 661 g (23 % / 77 %) — see §2                                                                                                                              | measured             |
| Chassis                 | LEGO Technic space-frame, 69-step build                                                                                                          | LEGO Technic frame, 84-step build ( <a href="#">instructions</a> ); 332 g bare                                                                                              | build file; measured |
| Printed parts           | TF-Luna holders, servo horn beam, motor clamp ( <a href="#">models/</a> )                                                                        | the same STL files                                                                                                                                                          | team                 |
| Steering                | single-servo Ackermann (MG90S)                                                                                                                   | non-Ackermann: an MG90 servo drives a vertical steering column that turns the front axle, 60° each side                                                                     | team                 |
| Wheelbase               | —                                                                                                                                                | 13 cm                                                                                                                                                                       | team                 |
| Wheels                  | front 41 mm, rear 55.6 mm                                                                                                                        | front 45 mm, rear 55 mm (rear tyres chosen for grip). Two 30 mm horizontal side rollers were fitted in the rebuild (build steps 71–76) and have since been removed — see §7 | team                 |
| Drive motor             | N20 gearmotor, 12 V, 600 rpm, via TB6612FNG                                                                                                      | MEX motor (Avishkaar), rated 6 V, 400 rpm, via TB6612FNG — see §3                                                                                                           | team                 |
| Gearing                 | 20T pinion → 28T crown on a LEGO differential (5:7)                                                                                              | unchanged                                                                                                                                                                   | team                 |
| Battery                 | 3S LiPo, 11.1 V, 2200 mAh                                                                                                                        | unchanged                                                                                                                                                                   | team                 |
| Power                   | pack → fast-charge module → Pi 5; pack → Buck-2 → 5 V rail (TF-Lunas, servo); TB6612FNG from the pack (2 — <a href="#">Power &amp; Sensors</a> ) | unchanged, plus the WS2812 LEDs on the 5 V rail; power switch on the battery lead; start button on the rear beams                                                           | team                 |
| Sensors                 | 3 × TF-Luna on a PCA9548A, BNO055, USB camera on a front arm at ~10 cm                                                                           | same sensors; the camera (Lenovo 300 FHD) now sits on top of a rear tower — see §4                                                                                          | photos; measured     |
| Lighting                | none                                                                                                                                             | 6 × WS2812 LEDs (three 2-LED strips) on the front frame — see §5                                                                                                            | team                 |

Wiring: [schemes/wiring\\_apac\\_2026.md](#). Photos: [v-photos/apac-2026-09-22/](#) — six views, 2026-09-22.

### 2. Weight and balance

| Quantity          | Value                                                 | Basis    |
|-------------------|-------------------------------------------------------|----------|
| Total, race-ready | 862 g                                                 | measured |
| Front / rear axle | 201 g / 661 g (23 % / 77 %); the two sum to the total | measured |
| Centre of mass    | 10.0 cm behind the front axle (13 cm × 661 / 862)     | derived  |
| Bare chassis      | 332 g, split 31 % / 69 %                              | measured |

| Quantity                        | Value                              | Basis   |
|---------------------------------|------------------------------------|---------|
| Everything added to the chassis | 530 g, about 82 % on the rear axle | derived |

Assuming the added parts share the chassis's split would have put 67 g too much on the front, which is why the full car was weighed per axle.

The driven rear wheels carry 77 % of the weight, which helps traction. The steered front wheels carry 23 %, on the smaller tyres, so front grip is what limits cornering. Planned check: measure the turning circle at walking pace and at race speed; a much wider circle at speed means the front is sliding.

### 3. Drive

- **Motor:** MEX motor (Avishkaar), rated 6 V and 400 rpm no-load (team).
- **Speed at the rated voltage:**  $400 \text{ rpm} \times 20/28 = 286 \text{ rpm}$  at the rear wheels;  $\times \pi \times 55 \text{ mm} = \mathbf{0.82 \text{ m/s no-load}}$  (derived).
- **Voltage the motor sees:** the TB6612FNG switches the pack (11.1 V nominal, 12.6 V full) with PWM, so the motor's average voltage is duty  $\times$  pack voltage. With the Nationals firmware, full commanded speed was 100 % duty (12.47 V measured at the motor terminals) and half speed was 48 % duty (5.95 V) — see 2a. On a 6 V motor, duty above about 48 % on a full pack averages more than its rating. The APAC firmware writes the duty directly with an 8-bit `analogWrite` and no cap: `SPEED = 210` is 82 % duty, 9.1 V average on the nominal pack and 10.4 V on a full one, 1.5–1.7 times the motor's 6 V rating (derived). Turning (100, 39 %), parking (70, 27 %) and the collision escape (80, 31 %) stay under the rating. The motor temperature after a full 3-minute run has not been measured yet.
- **Measured 2026-09-21** (pack at 12.3 V, `SPEED = 210`, so 10.1 V average; ammeter in the motor wire unless stated):

| Test                                             | Result                                                                                                     | Basis                          |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------|
| Rear wheels lifted, free-running                 | 0.33 A; rear wheel 422 rpm (slow-motion video of a taped wheel, counted over one minute) = <b>1.22 m/s</b> | measured                       |
| Driving on the mat                               | 0.599 A                                                                                                    | measured                       |
| Wheels held (stall), ammeter on the battery lead | 1.24 A, which is 1.51 A through the motor at 82 % duty                                                     | measured; motor figure derived |
| 3 m straight line from a standing start          | 3.36 s, 0.89 m/s on average with the launch included                                                       | measured                       |

- **Speed check.** Scaling the rated 400 rpm linearly to 10.1 V predicts 482 rpm at the rear wheel. The measured 422 rpm is 12.5 % lower, because linear scaling ignores the voltage lost in the winding and the drivetrain friction that shows up as the 0.33 A free-running current.
- **Motor model** (derived). Treating the motor as a standard DC motor (voltage = current  $\times$  resistance + speed  $\times$  motor constant) and fitting it to the free-running and stall points gives  $6.71 \Omega$  including the driver and  $0.128 \text{ V}\cdot\text{s}/\text{rad}$  at the gearmotor output. At the measured 0.599 A on the mat it predicts 0.94 m/s, and 3.37 s for the 3 m standing start (launch time constant 0.17 s). The measured time is 3.36 s.
- **Torque and grip** (derived; drivetrain efficiency beyond the free-running losses assumed 0.8). The rear axle gets about 0.04 N·m while cruising (1.4 N at the tyres, the rolling and drivetrain load on the mat) and 0.17 N·m at stall (6.2 N at the tyres). The rear tyres carry 6.5 N (661 g), so 6.2 N would need a friction coefficient of 0.95: a hard launch or a push against a wall spins the rear wheels on most surfaces before the motor stalls.

### 4. Camera placement

- **Mount:** lens centre  $24.8 \pm 0.5 \text{ cm}$  above the floor; optical axis  $12^\circ$  below horizontal ( $78^\circ$  from straight down); the hinge is glued at that angle (measured).
- **Field of view:** Lenovo lists a  $95^\circ$  diagonal field of view (product page), which is about  $56^\circ$  vertical and  $87^\circ$  horizontal at 16:9 (derived; not yet measured on this camera).

| Quantity                                      | Formula                                                           | Value                                                                                                         |
|-----------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Where the optical axis meets the floor        | $h / \tan \alpha$                                                 | 117 cm ahead of the lens; about $\pm 10$ cm per $1^\circ$ of angle error                                      |
| Nearest floor in view                         | $h / \tan(\alpha + \text{VFOV}/2)$                                | $\sim 29$ cm from the lens; the lens sits $\sim 10$ cm behind the front wheels, so $\sim 19$ cm ahead of them |
| Separation of pillar bases at 0.5 m and 1.0 m | $\text{atan}(h / 0.5 \text{ m}) - \text{atan}(h / 1.0 \text{ m})$ | $12.5^\circ$ ( $5.6^\circ$ with the Nationals 10 cm mount)                                                    |
| Tops of walls 1–3 m away                      | $\alpha - \text{atan}((h - 10 \text{ cm}) / D)$                   | $3.6^\circ$ – $9.2^\circ$ above the image centre                                                              |

$h$  = lens height,  $\alpha$  = pitch below horizontal, VFOV = vertical field of view,  $D$  = distance to the wall.

**Why the camera went up.** Reading range from the image depends on how low a pillar's base sits (the Nationals rule, 2 — Power & Sensors §3). Raising the lens from  $\sim 10$  cm to 24.8 cm makes that signal about 2.2 times stronger ( $12.5^\circ$  against  $5.6^\circ$  between pillars at 0.5 m and 1.0 m), and the centre of the image now lands  $\sim 1.2$  m ahead instead of 0.33–0.57 m. The separation keeps growing with height up to  $h = \sqrt{Z_{\text{near}} \cdot Z_{\text{far}}}$ , 55 cm for pillars at 0.3 m and 1.0 m, so the earlier statement in §3 of that document, that a higher mount compresses it, was wrong; it is corrected there.

### What it costs.

1. A blind zone: a pillar within  $\sim 19$  cm of the front wheels has its base below the frame, so the nearest-pillar rule needs a tie-break for boxes cut off at the bottom edge.
2. A view over the 100 mm walls: off-field objects can appear between the wall tops and the horizon, so the pillar search has to stay below the black wall band, not a fixed image row.

Planned checks: a tape on the floor slid until it just enters the bottom of the image (measures the blind zone directly); a start-position reference frame compared before each round (the hinge is glued, but this catches knocks).

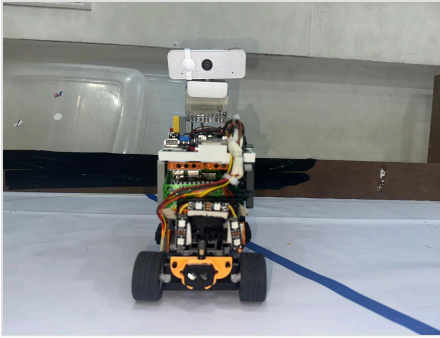
## Colour calibration

The colour thresholds are set on the competition mat during one practice round at each venue, under that venue's lighting (team). Two tools in <src/apac-2026/tools/> apply the same image processing as `main.py`: a contrast and brightness scale, a gamma curve, then contrast-limited histogram equalisation (CLAHE, clip limit 2.0,  $8 \times 8$  tiles) on the lightness channel.

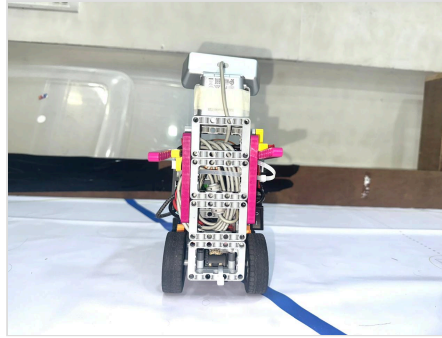
1. **Image.** `camera-settings.py` shows the live camera image with brightness, contrast and gamma sliders. The values chosen go into `BRIGHTNESS`, `CONTRAST` and `GAMMA` in `main.py` (committed: 0, 3.0 and 0.7).
2. **Colours.** `lab-calibration.py` converts the processed image to Lab and shows a mask while the lower and upper L, a and b sliders move. For each colour the sliders are set until only that colour shows in the mask, and the six values go into `COLOR_RANGES` in `main.py`.

Lab keeps lightness (L) apart from the two colour axes, so red (high a) and green (low a) can be separated with L left fully open, and a brighter or dimmer hall mostly changes L. The committed ranges do this: red is  $a \geq 184$  and  $b \geq 108$ , green is  $a \leq 101$ , both with L from 0 to 255. The blue and orange tape lines are found with fixed HSV ranges (`COURSE_LINE_HSV_RANGES` in `main.py`), not with these tools.

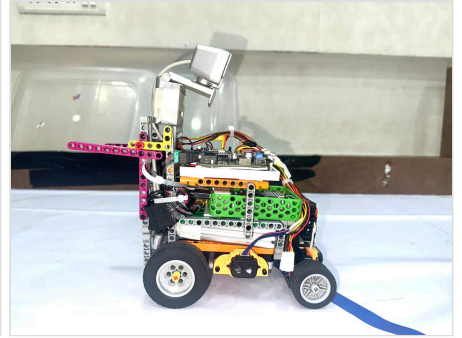
## The six views



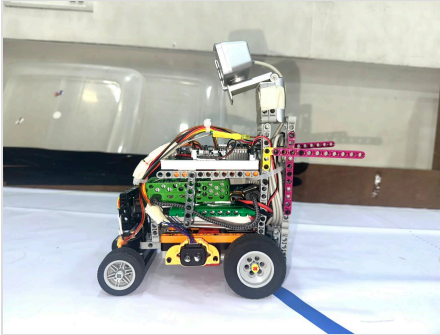
Front



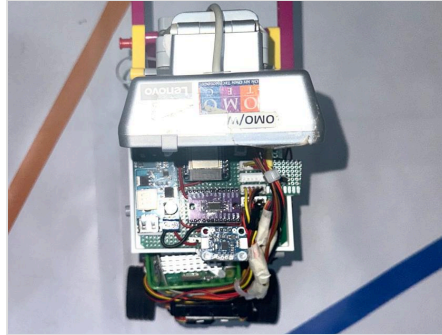
Rear



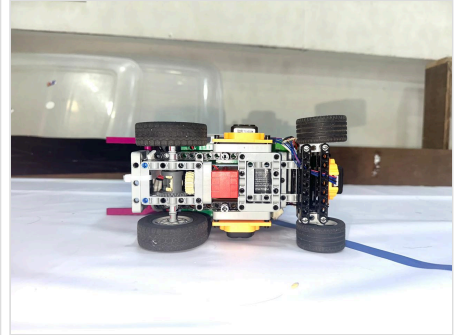
Left



Right



Top



Bottom (laid on its side)

## 03 Power and sensing

*What powers the car, what it senses with, and what can fail (criterion 2).*

### 5. Lighting and power additions

- **LEDs:** 6 × WS2812 on the front frame; data from ESP32 GPIO4; power from the 5 V Buck-2 rail (team). The firmware lights them white at `setBrightness(50)`, about 20 %: roughly 12 mA per LED and 0.07 A for all six by the 60 mA-per-LED full-white rule of thumb in the [Adafruit NeoPixel Überguide](#) (derived). Full white would be 0.36 A.
- **Switch and button:** one power switch on the battery lead; the start button on the rear beams (team). The rules require exactly one power switch and one start button (2 — [Power & Sensors §6](#)).

### Power budget, measured 2026-09-21

Each branch was measured on its own lead from the pack (ammeter and voltmeter, pack at 12.3 V). Pack currents for the two 5 V converters are given at the nominal 11.1 V, the conservative case used in [2a](#), with the converters assumed 88 % efficient; the motor branch is as measured.

| Branch                   | Feeds                    | Basis                           | Idle          | Driving                      | Worst case     |
|--------------------------|--------------------------|---------------------------------|---------------|------------------------------|----------------|
| Buck-2 (5 V)             | 3 TF-Luna, servo, 6 LEDs | measured at its input           | 0.13 A        | 0.23–0.32 A (servo sweeping) | 0.32 A         |
| Fast-charge module (5 V) | Pi 5, camera, ESP32      | measured peak, 11.8 W at the Pi | 1.21 A        | 1.21 A                       | 1.21 A         |
| TB6612FNG                | drive motor              | measured                        | 0             | 0.49 A (0.599 A × 82 % duty) | 1.24 A (stall) |
| <b>Pack total</b>        |                          |                                 | <b>1.34 A</b> | <b>1.93–2.02 A</b>           | <b>2.77 A</b>  |

With the Pi's peak treated as its average, a 2200 mAh pack used to 80 % gives about 52 minutes of driving, about 17 three-minute runs (derived). The prediction for the Nationals car in [2a](#) was 0.95–1.40 A driving; the difference is the Pi (an 11.8 W peak against 6.2–8.5 W predicted as an average) and the larger motor.

| Part                          | Load                                               | Rating (vendor)              | Use                                  |
|-------------------------------|----------------------------------------------------|------------------------------|--------------------------------------|
| TB6612FNG, one channel        | 1.51 A through the bridge at stall; 0.60 A driving | 1.2 A continuous, 3.2 A peak | <b>126 % at stall</b> ; 50 % driving |
| Pi 5 USB ports without USB-PD | camera + ESP32                                     | 600 mA in total              | see §6                               |
| Battery, 3S 2200 mAh 60C      | 2.77 A worst case                                  | 132 A                        | 2 %                                  |

### 6. New failure points

| Risk                                                 | Effect                                                                                                                                                                                                       | Check or mitigation                                                                                            |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Pi 5 fed from a USB-A fast-charge module (no USB-PD) | The Pi peaked at 11.8 W (2.36 A at 5 V, measured) and the module's rating has not been read; without USB-PD the Pi limits its USB ports to 600 mA in total, and the camera and the ESP32 both draw from them | Run <code>vcgencmd get_throttled</code> after a full run; move to a 5 A USB-C PD supply if it reports problems |
| WS2812 data at the ESP32's 3.3 V                     | The LED input-high threshold is $0.7 \times V_{DD}$ (3.5 V at 5 V); a marginal signal can flicker or show the wrong colour, tinting the scene                                                                | Watch for flicker in the camera image; add a level shifter if seen                                             |
| 6 V motor on the 11.1 V pack                         | Heat and brush wear: <code>SPEED = 210</code> averages 9.1–10.4 V, above the 6 V rating (§3)                                                                                                                 | Temperature check after a 3-minute run                                                                         |

| Risk                                                                                                                     | Effect                                                                                                                                                 | Check or mitigation                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drive motor held against a wall                                                                                          | 1.51 A through the TB6612FNG at stall, above its 1.2 A continuous rating (§5); the driver's thermal shutdown or the motor's heating would stop the car | The Pi's distance logic backs away (escape at 31 % duty for 0.30 s, at least 3 cm of progress, at most 2 attempts); on most surfaces the rear wheels spin before a full stall (§3) |
| Firmware front stop: the ESP32 stopped the drive motor when the front TF-Luna read under 18 cm or its reading went stale | <b>Observed:</b> it stopped the car and ended a run (team)                                                                                             | Removed from the firmware on 2026-09-10; collision handling moved to the Pi (§7)                                                                                                   |
| Camera angle                                                                                                             | Pixel-based settings shift if the angle moves                                                                                                          | Hinge glued; reference frame before each round                                                                                                                                     |
| Background above the walls                                                                                               | Off-field red or green objects read as pillars                                                                                                         | Search only below the wall band                                                                                                                                                    |
| Blind zone ahead of the wheels                                                                                           | The nearest-pillar rule ties for boxes cut off at the bottom                                                                                           | Tie-break rule in the code                                                                                                                                                         |

## 8. ESP32 pin map

From the Nationals firmware ([round2\\_ino.ino](#)) and the 2026-09-20 wiring draft; only GPIO4 is new. Checked against the APAC firmware ([src/apac-2026/](#)) on 2026-09-21: it reads the centre TF-Luna on mux channel 3 (channel 1 at Nationals).

| Signal                                                                                                           | GPIO              |
|------------------------------------------------------------------------------------------------------------------|-------------------|
| I <sup>2</sup> C SDA / SCL — PCA9548A at 0x70: ch0 left, ch3 centre, ch2 right TF-Luna (0x10), ch4 BNO055 (0x28) | 21 / 22           |
| Steering servo                                                                                                   | 13                |
| TB6612FNG AIN1 / AIN2 / PWMA / STBY                                                                              | 25 / 26 / 33 / 27 |
| WS2812 data                                                                                                      | 4                 |
| Start button                                                                                                     | 32                |

## 04 Software and obstacle strategy

*How main.py drives the Obstacle Challenge and how the car parks (criterion 3).*

**Scope.** How the Obstacle Challenge code works. §1-§6 describe `main.py` as released in v0.4.0 (from the coach's commit of 2026-09-08; [that file](#)); §10 lists what the version now in the repository, dated 2026-09-18, changes. §8 covers the parking modules and §9 the pillar and line detection of the current file. Names in capitals are states; names in `code` are constants, given with their values. The Nationals software is in [3 - Software](#) and the vehicle in [APAC 2026 vehicle](#).

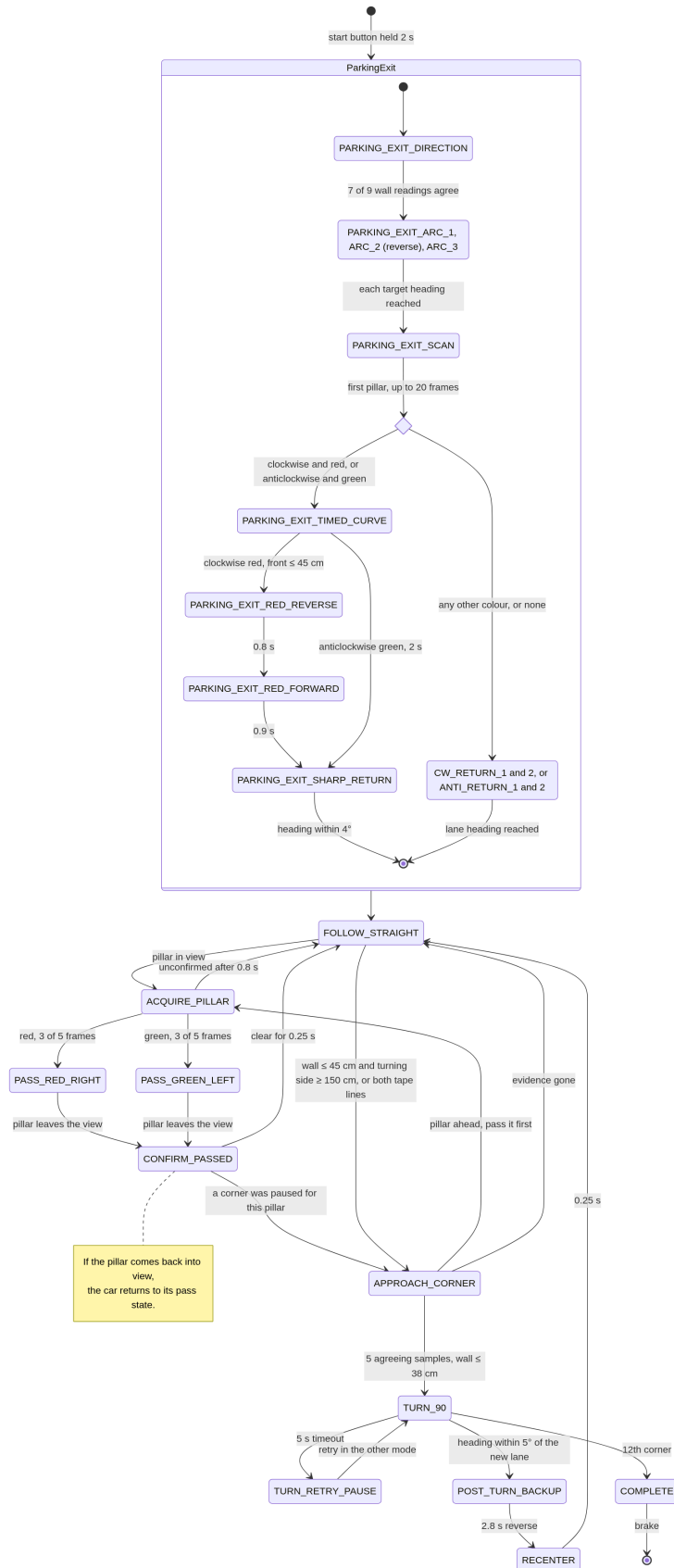
### 1. The control loop

In a loop that sleeps 10 ms between passes (`NAVIGATION_LOOP_SECONDS`), the Raspberry Pi:

1. reads the newest ESP32 telemetry line: heading, the left, centre and right TF-Luna distances, and a validity mask;
2. takes the newest camera frame and runs the pillar and tape-line detector (`UnifiedPillarDetector`, §9);
3. calls the controller (`ObstacleChallengeController.update`), which sends one `speed,direction,steer` command to the ESP32.

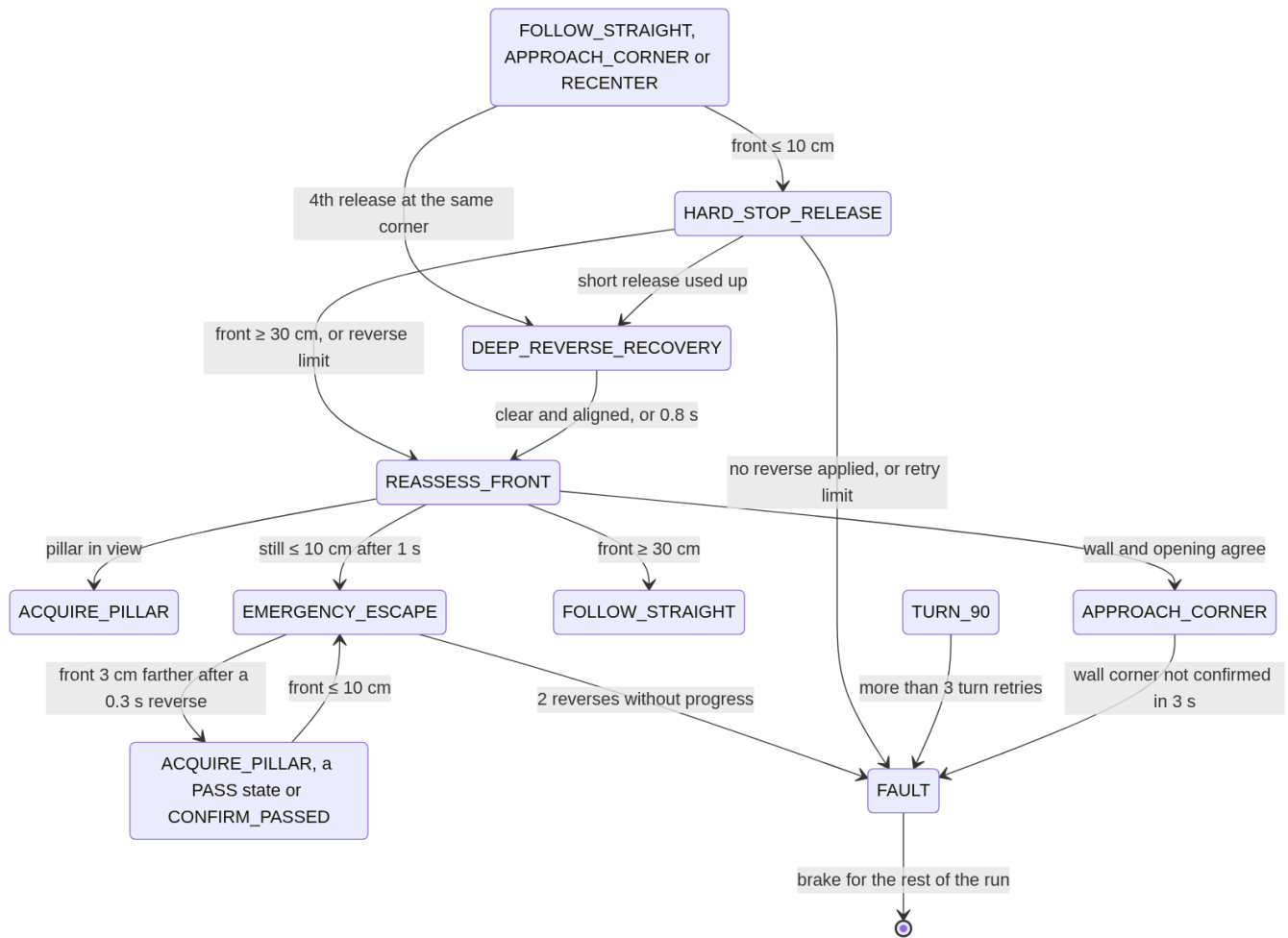
The run starts once the start button has been held for 2 s: the ESP32 then sends `OK` and the Pi answers `OK`. It ends in `COMPLETE` after 12 corners (`COUNTER_MAX`), that is three laps. Parking exit is on (`ENABLE_PARKING_EXIT = True`); parking at the end is off (`ENABLE_PARKING_IN = False`). Every state change is printed as `[NAV] OLD -> NEW: reason`.

## 2. The run



- Red pillars are passed on the right and green ones on the left.
- From any pillar state, a pillar seen beyond both tape lines within 5 s of a pass (`ALPHA_POSITION_WINDOW_SECONDS`) belongs to the next straight, so the car goes to `APPROACH_CORNER` and takes the corner first.
- Every driving state brakes and holds while the telemetry is older than 0.30 s (`TELEMETRY_STALE_SECONDS`), a TF-Luna or the heading is invalid, or the camera frame is older than 0.30 s (`CAMERA_STALE_SECONDS`). It resumes in the same state, and the hold does not count against that state's timers.

### 3. Recovery and faults



EMERGENCY\_ESCAPE resumes whichever state it interrupted. The parking-exit states also go to FAULT when a timed step overruns (4 s to reach the first red pillar, 5 s to reach the lane heading).

### 4. Why the states are split this way

| State                  | Reason                                                                                                                                                                                                                                                                                                        |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PARKING_EXIT_DIRECTION | The lane direction must be known before the first corner. The code takes it from which side wall is nearer: 7 of the last 9 readings must agree, each with left and right at least 15 cm apart (DIRECTION_MIN_DIFFERENCE_CM).                                                                                 |
| PARKING_EXIT_SCAN      | A pillar may stand just outside the parking lot. If it is on the side the car turns toward (red when driving clockwise, green anticlockwise), a timed pass keeps the car on its correct side; otherwise two heading arcs return the car to its lane.                                                          |
| ACQUIRE_PILLAR         | One frame of red or green must not steer the car, so a pillar is followed only once 3 of 5 frames confirm it.                                                                                                                                                                                                 |
| CONFIRM_PASSED         | The camera loses a pillar before the rear of the car has cleared it, so the car holds its line for 0.25 s (PILLAR_CLEARANCE_HOLD_SECONDS).                                                                                                                                                                    |
| APPROACH_CORNER        | Neither a tape line nor a short front distance alone proves a corner. A turn needs the end wall within 38 cm (CORNER_TRIGGER_CM), the side the car turns toward open to at least 150 cm (OBSTACLE_CORNER_SIDE_OPEN_CM) and the heading within 8° of the straight, in 5 agreeing samples over at least 0.30 s. |
| TURN_90                | The target is the straight's heading plus or minus 90°, reached within 5° (TURN_HEADING_TOLERANCE). When the outer side has at least 60 cm of room (CORNER_REVERSE_CLEARANCE_CM), the turn starts with a reverse arc of up to 0.8 s; otherwise it is driven forward.                                          |
| POST_TURN_BACKUP       | After each corner the car reverses for 2.8 s (POST_TURN_BACKUP_SECONDS) so that the next straight and its pillars fit in the camera's view.                                                                                                                                                                   |

## 5. Edge cases the code handles

| Situation                                        | What happens                                                                                       |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Telemetry or camera stale, or a sensor invalid   | Brake and hold the current state; resume when fresh.                                               |
| The same pillar seen again just after a pass     | Ignored for 0.6 s ( <code>PILLAR_REACQUIRE_COOLDOWN_SECONDS</code> ).                              |
| Candidate pillar not confirmed                   | Back to FOLLOW_STRAIGHT after 0.8 s ( <code>PILLAR_ACQUIRE_TIMEOUT_SECONDS</code> ).               |
| Pillar ahead during a corner approach            | The corner is paused, the pillar is passed, then the corner approach resumes.                      |
| Pillar beyond the corner seen right after a pass | The corner is taken first (§2).                                                                    |
| Front within 10 cm                               | A short reverse, then a fresh check for a pillar, a corner or a clear front before moving on (§3). |
| Repeated releases at one corner                  | After the third, a longer reverse at speed 55 for up to 0.8 s, aiming for 70 cm of clearance.      |
| Turn not finished in 5 s                         | Retry with the other turn mode (forward or reverse), up to 3 retries.                              |
| 12th corner                                      | COMPLETE: the car brakes and stays stopped.                                                        |

## 6. Known limits

- The driving direction is locked at the start and does not change during the run.
- The last-resort front trigger ( `PYTHON_EMERGENCY_RELEASE_CM = 10` ) is inside the TF-Luna's blind zone: its specified range starts at 0.2 m (vendor). Corners are normally handled from 45 cm, so this trigger only acts after a misjudgement.
- Since the firmware stop was removed on 2026-09-10, the ESP32 always reports `hardStop = 0`, so the `hard_stop` checks in `main.py` never fire; some comments in `main.py` still describe an ESP32 stop at 18 cm.
- With parking at the end off, `PARKING_ENTRY_SEARCH` and the `PARKING_IN_` states are never reached. No transition leads into the `PARKING_EXIT_CW_RED_` and `PARKING_EXIT_ANTI_GREEN_` states.
- `FAULT` stops the car for the rest of the run.
- `main.py` as committed does not call the parking modules (§8). The docstring of `partial_parking.py` says `main.py` "can hand control to `partial_parking_main` after corner 12"; no such call is in the committed `main.py`.
- Both parking modules end the entry when the front TF-Luna reads 5 cm or less ( `ENTRY_STOP_FRONT_CM = 5` ), and `partial_parking.py` also uses limits of 6, 8 and 15 cm ( `REVERSE_BLOCK_SEARCH_SIDE_STOP_CM`, `REPOSITION_SIDE_CLEARANCE_CM`, `REPOSITION_FRONT_CLEARANCE_CM` ). All are inside the TF-Luna's blind zone: its specified range starts at 0.2 m (vendor).
- In `parallel_parking.py`, `FINAL_REVERSE`, `FINAL_BRAKE` and `FINAL_FORWARD` are defined, but no transition leads into `FINAL_REVERSE`, so none of the three is reached; `PARALLEL_ALIGN` ends the parking in `COMPLETE`.
- The docstrings of `parallel_parking.py` say its detector does not use the blocks' colour; the code ( `ParkingGeometryDetector.extract` ) keeps only magenta pixels.

## 7. Validation

The metrics that exist, and how the next ones are collected.

- **Detector, per frame.** The live view shows the tracked pillar's confidence, hits out of 5, bottom position and area band (§9); in the two captured frames of 2026-09-12 the track had 5 of 5 hits at confidence 0.92 and 0.86.
- **Nationals stack.** The test levels T1-T7 in `tests.md` (data integrity, model evaluation, operating-point sweep, deploy parity, on-target latency, firmware bench, track), with what was run and what was not, and the track recordings in `other/test-runs-2026-08-08` and `other/test-runs-2026-08-09`.
- **APAC bench, 2026-09-21.** Drive and power measurements in [vehicle §3](#) and [§5](#): free-running and stall currents, 422 rpm at the rear wheel, 0.89 m/s over 3 m from a standing start, the current of each supply branch.
- **Run logs exist on the car, not here yet.** The 2026-09-18 `main.py` writes `run_logs/round2_run_<date_time>.csv` beside itself on every run ( `ENABLE_RUN_CSV_LOGGING = True` ): one row per telemetry sample and one per event, with lap, corner count, pillars this straight, state, heading and target, the three distances, the command sent and applied, the tracked pillar with its band, and the tape depths. Those files give corners completed, pillars passed,

recoveries and faults per run; none is in this repository yet. For parking, `parallel_parking.py` with `ENABLE_RUN_LOGGING = True` (committed as `False`) writes a CSV and an annotated video per attempt (§8).

## 8. Parking modules

Two files beside `main.py`, added on 2026-09-23, park the car at the end of the run. `main.py` does not call them: v0.4.0 had parking at the end off, and the 2026-09-18 version parks with its own routine (§10): each runs on its own and imports `main.py` for the camera, the serial link, telemetry, the live view and the shutdown. Constants named in this section are in the parking files.

| File                             | What it does                                                                                                                                                                     | Start                                                                            |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <code>parallel_parking.py</code> | Parallel parking ( <code>BUILD_ID = "slot-guided-entry-v29"</code> ): finds the two parking-lot limitations with the camera, drives into the gap between them and straightens up | <code>python parallel_parking.py --direction clockwise --last-pillar auto</code> |
| <code>partial_parking.py</code>  | Direct entry: the same approach and block search, then a drive into the space on a locked heading ("Standalone 90 degree parking" in its help text)                              | <code>python partial_parking.py --direction clockwise --last-pillar auto</code>  |

`--direction` is `clockwise` or `anticlockwise`; `--last-pillar` is `auto` (detect a red or green final pillar), `red`, `green` or `none`. Both wait for the start signal as `main.py` does, and end in `COMPLETE` with the brake held (exit code 0) or in `FAULT` with a printed reason (exit code 1).

**At APAC the team parks with `parallel_parking.py`** (decided 2026-09-23). A full parallel park scores 15 points against 7 for a partial or non-parallel one (2026 General Rules, scoring element 1.8.2), and touching a limitation ends the round (rule 9.24.7): the entry is deliberately slow (speed 40, a "slow, controlled approach" in the code) and the reverse strokes are capped so the car cannot back out. The `main.py` that hands over to it after the twelfth corner is not yet in this repository; `partial_parking.py` is kept as the fallback. The `main.py` now in the repository (2026-09-18) parks with its own routine instead (§10). Every `main.py` name the parking modules use also exists in the 2026-09-18 file.

**Camera calibration.** Both files need `parking_camera_calibration.json` beside them. It depends on how the camera is mounted, so it is made on the vehicle and is not in the repository: `python parallel_parking.py --calibrate` shows the camera image, and clicking the four corners of a rectangle on the floor (near-left, near-right, far-right, far-left; 100 × 150 cm unless `--calibration-width-cm` and `--calibration-depth-cm` say otherwise) saves the homography from image to floor. If the file is missing, `parallel_parking.py` starts this calibration itself and exits after saving.

### Finding the space (`parallel_parking.py`)

- Colour mask.** Pixels that are magenta in both HSV (`MAGENTA_HSV_LOW` 135, 65, 45 to `MAGENTA_HSV_HIGH` 179, 255, 255) and Lab ( $a \geq 145$ ,  $b \leq 175$ ). The top 28 % of the image is ignored (`ROI_TOP_FRACTION = 0.28`); one opening and two closings with a 5 × 5 kernel remove specks and fill gaps.
- Shape.** Each contour's rotated rectangle must fit a 200 × 20 × 100 mm limitation seen face-on, edge-on or at an angle: aspect 1.15 to 6, rectangularity at least 0.42, magenta fill at least 0.45, width at most 30 % and height 5.5 to 72 % of the image, lower edge below 55 % of the image height and not cut off by the bottom of the image.
- Floor position.** The midpoint of the rectangle's two lowest corners goes through the calibration homography, which gives the block's position on the floor in cm.
- Two blocks.** A block is confirmed after 3 hits in its last 6 frames (`BOUNDARY_REQUIRED_HITS`, `BOUNDARY_HISTORY`). Block 1 is the best-scoring rectangle on the parking side of the image. Block 2 must not overlap it, and must be 25 to 120 cm from it on the floor, within 55 cm of it sideways and at least 15 cm farther away.
- Slot.** The slot is the floor midpoint of the two blocks. It is confident when both blocks are confirmed, neither has been missed for more than 2 frames, they are 25 to 120 cm apart and both have a confidence of at least 0.55. Its bearing from the camera steers the car (`SLOT_BEARING_KP = 2.0`) and is used only within  $\pm 30^\circ$  (`SLOT_MAX_USABLE_BEARING_DEG`).

## States (parallel\_parking.py)

| State                 | What the car does                                                                                                                                                                                                                                                                                                                                                 | Leaves when                                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEARCH_LAST_PILLAR    | Looks for the final pillar with <code>main.py</code> 's pillar detector                                                                                                                                                                                                                                                                                           | a pillar is confirmed: <code>PASS_LAST_PILLAR</code> ; no pillar and block 1 confirmed: <code>TRACK_OPENING</code>                                                                                              |
| PASS_LAST_PILLAR      | Steers past the final pillar                                                                                                                                                                                                                                                                                                                                      | red path: <code>SEARCH_FIRST_BOUNDARY</code> ; green path: <code>GREEN_RETURN_STRAIGHT</code> back to the start heading, <code>GREEN_FORWARD_STRAIGHT</code> for 1.0 s, then <code>SEARCH_FIRST_BOUNDARY</code> |
| SEARCH_FIRST_BOUNDARY | Drives at speed 40 on the start heading turned 45° towards the parking side (5° after a green-path pillar). If block 1 is not confirmed within 4 s, a short reverse (0.6 s at speed 55, steer 35) and the search restarts; each retry adds 5° of offset (up to 55°), 0.2 s of search time and a longer, faster, sharper reverse (up to 1.0 s, speed 75, steer 55) | block 1 confirmed: <code>TRACK_OPENING</code>                                                                                                                                                                   |
| TRACK_OPENING         | The same search                                                                                                                                                                                                                                                                                                                                                   | block 2 confirmed and the slot confident: <code>ALIGN_VIRTUAL_CUTOUT</code>                                                                                                                                     |
| ALIGN_VIRTUAL_CUTOUT  | Speed 40, steering on the slot bearing (at most 25)                                                                                                                                                                                                                                                                                                               | 8 consecutive updates with a usable bearing ( <code>ALIGN_CONFIRM_SAMPLES</code> ): <code>ENTER_SPACE</code> ; front TF-Luna at 5 cm or less: <code>PARALLEL_ALIGN</code>                                       |
| ENTER_SPACE           | Speed 40, steering on the slot bearing (at most 12)                                                                                                                                                                                                                                                                                                               | front TF-Luna at 5 cm or less: <code>PARALLEL_ALIGN</code>                                                                                                                                                      |
| PARALLEL_ALIGN        | Speed 35 at full correction steer (45): reverse strokes of 0.8 s, then forward until the front TF-Luna reads 5 cm                                                                                                                                                                                                                                                 | heading within 2° of 0° on 3 consecutive updates: <code>COMPLETE</code>                                                                                                                                         |

The red and green paths swap in an anticlockwise run (`pillar_behavior`). After a green-path pillar, each block search ends in `FAULT` after 15 s (`GREEN_PARKING_SEARCH_MAX_SECONDS`), and the return to the start heading after 10 s.

## What `partial_parking.py` changes

- **No final pillar.** With `--last-pillar none`, or when no final pillar is confirmed within 3 s, it drives straight for 2 s at speed 80 (`NO_PILLAR_STRAIGHT`), then searches for block 1 in reverse at speed 50 (`REVERSE_BLOCK_SEARCH`).
- **Green path.** Clockwise with a green final pillar (anticlockwise with a red one) it steers 45° forward until the centre TF-Luna reads 30 cm, then turns slowly at full lock the other way to a set heading before the block search (`GREEN_CW_TURN_RIGHT`, `GREEN_CW_TURN_LEFT`; mirrored as `RED_ACW_TURN_LEFT`, `RED_ACW_TURN_RIGHT`).
- **Opening check.** Where the parallel module would start aligning, it requires a visible opening at least 6.5 % of the image wide (`MIN_VISIBLE_ENTRY_GAP_RATIO`) and a slot at least 20 cm ahead with a valid bearing. A narrower opening gets one repositioning (`REPOSITION_AWAY`, `REPOSITION_FORWARD`, `REPOSITION_RETURN`) before a `FAULT`.
- **Entry.** It locks the entry heading at the current heading plus the slot bearing (at most  $\pm 30^\circ$ ) and drives in at speed 40 (`ENTER_SPACE`). Steering is  $2.0 \times$  the heading error plus  $0.6 \times$  its integral, which builds only once the slot is closer than 40 cm, plus a smoothed term of at most 8 that centres the opening in the image while the slot is 40 cm or more away; at most 24 in total.
- **Stop.** At a front reading of 5 cm or less, or when the front reading is missing or stale, it brakes (`FRONT_SENSOR_HOLD`): 3 fresh readings at the stop move on to `PARALLEL_ALIGN`, and 3 fresh clear readings resume the previous state. `PARALLEL_ALIGN` works as in the parallel module but counts only fresh telemetry.

## Why it is built this way

Reasons as the code's comments give them.

| Choice                                                                                                    | Reason given in the code                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The TF-Luna sensors are not used while searching for or aligning with the blocks                          | The blocks' geometry comes from the full camera view; distance matters only once the car is in the space                                                                       |
| A short camera miss does not brake the car                                                                | The brake is not pulsed between camera frames: the selected motion continues until vision recovers, and during alignment the car drives straight and restarts the confirmation |
| Avoiding the final pillar starts on its first sighting, but the pass needs several frames of confirmation | At driving speed a one-frame delay can put the pillar past the useful steering window; a false detection can steer briefly but cannot latch the pass                           |
| The block search holds a heading offset towards the parking side                                          | So that the blocks enter the camera view; an earlier path turned into the parking area before block 1 was confirmed                                                            |
| Reverse strokes in PARALLEL_ALIGN stop after 0.8 s; forward moves stop on the front TF-Luna               | So that the car cannot back out of the space                                                                                                                                   |
| The shape limits are broad                                                                                | A limitation can be seen edge-on, face-on, at an angle or cut off by the image edge                                                                                            |
| One magenta contour never updates both blocks, and a sudden inward jump is taken as the next block        | Block 1 leaves the image through its left edge in a clockwise run (the right edge anticlockwise), so a new rectangle further in is block 2, never a new sample of block 1      |
| <code>partial_parking.py</code> ignores a pillar cut off by the image edge                                | Such an object belongs to the next lane, not to the final pillar ahead                                                                                                         |

## Failure exits

Every FAULT prints its reason and holds the brake.

- `parallel_parking.py`: after a green-path pillar, the return to the start heading takes more than 10 s, or block 1 or block 2 is not found within 15 s.
- `partial_parking.py`: no valid IMU heading when the entry heading is locked; the opening is still narrower than 6.5 % of the image after one repositioning; the slot is not ahead with a valid bearing; something in front while the final pillar is classified, during the no-pillar straight or during the pillar pass; the pillar pass takes more than 4 s; too little side clearance in the reverse search; too little front or side clearance while repositioning; the opening is lost before the car passes it.

## 9. Pillar and line detection

What `UnifiedPillarDetector` does with each  $1280 \times 720$  frame, and why. Constants are in `main.py` (the 2026-09-18 file; v0.4.0 values in brackets where they differ); the two figures at the end show the result on the mat.

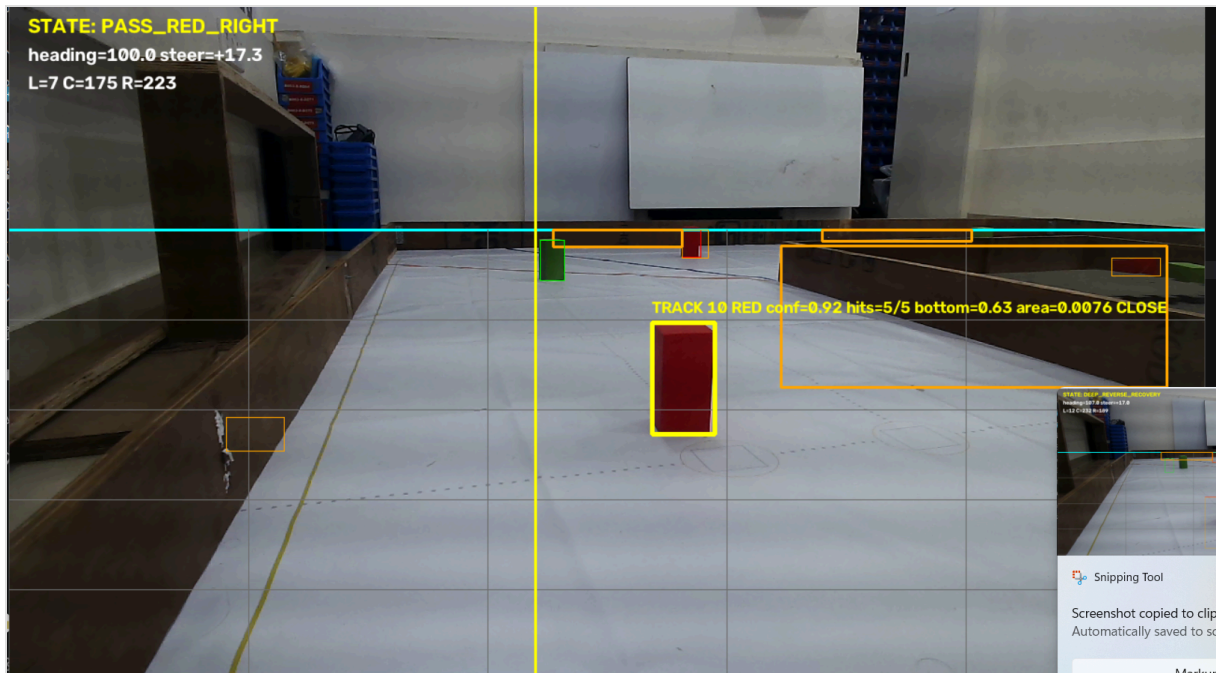
1. **Image preparation.** The frame is stretched (`CONTRAST = 3.0`, `BRIGHTNESS = 0`) and gamma-corrected (`GAMMA = 0.7`), then converted to Lab; CLAHE (clip 2.0,  $8 \times 8$  tiles) evens out the lightness channel. Lab keeps lightness apart from the two colour axes, so red (high a) and green (low a) separate with L left open, and a brighter or darker hall mostly moves L ([vehicle §4](#)). The venue's thresholds come from one practice round with `lab-calibration.py`.
2. **Central view.** Before detection a strip on each side of the frame is set to neutral grey (128, outside every Lab range) without moving pixel coordinates: 16 % of the width on each side (`CAMERA_SIDE_MASK_FRACTION`), 25 % for 5 s after a corner (`POST_CORNER_VIEW_SIDE_FRACTION`, `POST_CORNER_NARROW_VIEW_SECONDS`), none after the twelfth corner, so parking sees the whole frame. Not in v0.4.0.
3. **Colour masks.** One Lab range per colour (`COLOR_RANGES`, as committed): red a at least 184 and b at least 108; green a at most 101; blue a 153 to 186 and b at most 90; orange a at least 130 and b at least 150; L unconstrained. Each mask is opened once and closed twice with a  $3 \times 3$  kernel, and the top third of the image is cleared (`DETECTOR_ROI_TOP_RATIO`): from the 24.8 cm mount that third lies above the tops of the walls out to about 3 m ([vehicle §4](#)), so it holds only the hall.
4. **Candidate shapes.** Each contour is tested by size and shape. Pillars (red, green): area at least 0.045 % of the frame and height at least 2.5 % of the frame height; height to width 0.65 to 8 (a  $50 \times 50 \times 100$  mm sign is about twice as tall as wide; the range allows partial views and perspective); rectangularity at least 0.35 and solidity at least 0.60, since a pillar is a solid rectangle and tape edges and reflections are not. Tape lines (blue, orange):

area at least 0.03 %, height at least 1 %, height to width 0.2 to 10, rectangularity at least 0.25, solidity at least 0.50. Confidence =  $0.20 \times \text{area} + 0.20 \times \text{height} + 0.25 \times \text{rectangularity} + 0.35 \times \text{solidity}$ , with area and height capped at  $5 \times$  the minimum area and 25 % of the frame height; candidates under 0.45 are dropped (`DETECTOR_MIN_CONFIDENCE`).

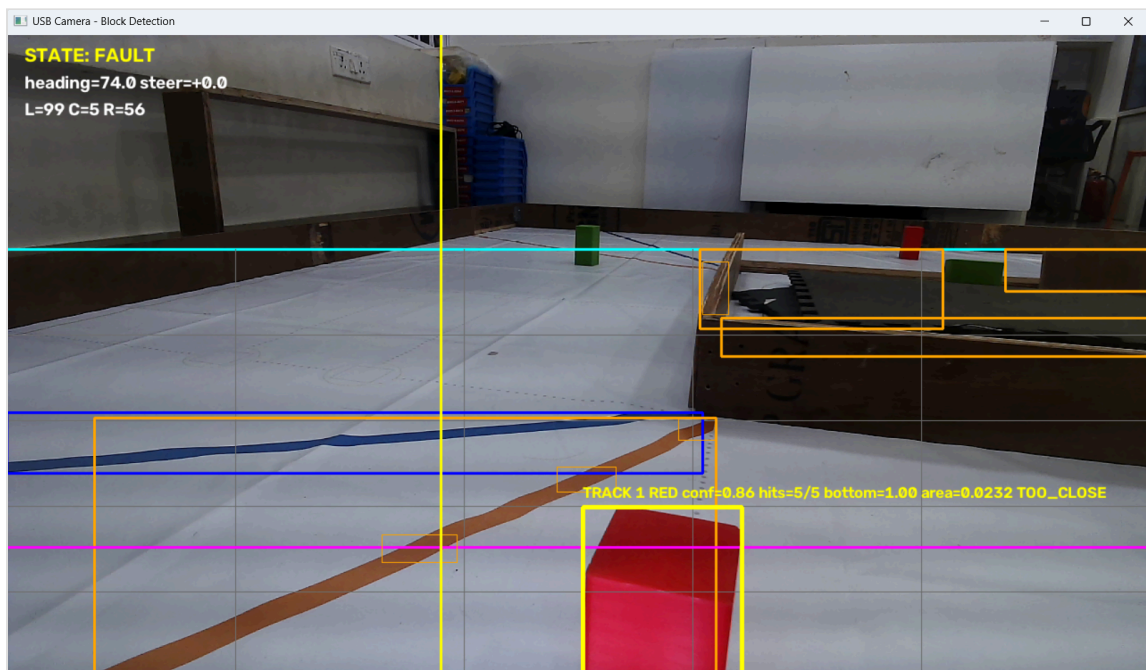
5. **Tape lines and the corner reference.** The mat's blue and orange lines mark the approach to a corner. They are found in HSV (blue hue 95-135, orange hue 3-22) below the top third, with the same morphology; a line must cover 0.025 % of the frame, span 10 % of its width and be at least twice as wide as tall. The lowest line's mid-height is its depth; when both colours are visible, the deeper of the two is the corner reference (the magenta line in the live view).
6. **Which pillar counts.** With the corner reference visible (and not suppressed for 3 s after a turn, `POST_TURN_TAPE_IGNORE_GRACE_SECONDS`), a pillar whose centre is above the reference, farther away than the tape, is ignored when it is entirely above it or sits in an outer column of the  $5 \times 5$  grid: it stands beyond the corner or in a side lane (`IGNORE CORNER` in the live view). Two rules keep the car from dropping the obstacle in front of it: the track the controller is already passing keeps priority even when tape moves in front of it (`FRONT PRIORITY`), and when a larger, lower interior pillar and a smaller outer pillar are both beyond the reference, the interior pillar stays actionable if it is  $1.25 \times$  larger (`CORNER_FOREGROUND_AREA_ADVANTAGE`). A small ignored pillar never suppresses a valid foreground pillar.
7. **Tracking.** One pillar is tracked at a time. A new track starts on the best candidate by  $0.55 \times \text{bottom} + 0.35 \times \text{confidence} + 0.10 \times \text{area}$ : the lowest in the image is the nearest. Each frame the track is matched to a same-colour candidate by centre distance (at most 0.25 of the frame) or overlap (IoU at least 0.05), and its position is smoothed (0.65 new, 0.35 old). A track is confirmed after 3 hits in its last 5 frames (`DETECTOR_CONFIRMATION_HITS`, `DETECTOR_TRACK_HISTORY`); an unconfirmed track is dropped after 2 misses (`DETECTOR_UNCONFIRMED_MISSES`); a confirmed track stays locked until the controller records the pass, so a brief occlusion cannot switch the target. Confidence is  $0.75 \times \text{the detection's confidence} + 0.25 \times \text{hits}/5$ , and decays by 10 % per missed frame.
8. **Two pillars per straight** (2026-09-18 file). At most two pillars are taken on each straight (`MAX_PILLARS_PER_STRAIGHT = 2`); once both are passed, no new pillar is taken until the corner (live view: `CORNER NEXT - NO NEW PILLAR`). One high track in the next lane is deferred rather than its whole colour suppressed, and right after a turn nearby pillars take priority (`defer_background_candidate`). A wide, low rectangle, at least twice as wide as tall, is treated as a parking-lot limitation, not a pillar; in laps 1 and 2 the car steers gently around it (speed 60, steering 15).
9. **Range and control.** The pillar's distance band comes from its area: far under 3,200 px×px (4,000), near under 7,000, close under 20,000, otherwise too close; the  $1280 \times 720$  breakpoints are kept as ratios of the frame (`PILLAR_AREA_NEAR_RATIO`, `PILLAR_AREA_CLOSE_RATIO`, `PILLAR_AREA_TOO_CLOSE_RATIO`). Each band selects a speed, gain and steering limit (`PILLAR_DISTANCE_PROFILES`; far: speed 100, gain 0.70, 3 to 12 of steer (5 to 18); too close: speed 80, gain 1.50, 38 to 52). The controller steers the pillar to a target column, red to 38 % of the width (25 %) so the car passes on its right and green to 62 % so it passes on its left (`PILLAR_RED_TARGET_X`, `PILLAR_GREEN_TARGET_X`), with gain 200 (`PILLAR_LATERAL_KP`), at most 40 of correction and smoothing 0.65.

**The live view** (`annotate`): the cyan line is the top of the search area; the grey grid is the  $5 \times 5$  columns and rows; blue and orange boxes are tape lines; the magenta line is the corner reference; thin boxes are candidates; a magenta box is a pillar ignored as beyond the corner (`IGNORE CORNER Cn`,  $n = \text{grid column}$ ); a yellow box marks the pillar being passed (`FRONT PRIORITY`); the tracked pillar's box is yellow once confirmed and cyan before, labelled `TRACK id COLOUR conf hits/5 bottom area BAND`, with a vertical line at its target column; the text lines give the state, heading and steer, and the three TF-Luna readings in cm. The 2026-09-18 file adds `PILLAR SLOTS n/2`, `CORNER NEXT - NO NEW PILLAR`, `SIDE-SENSOR SLOT 2 IN n s`, and two cyan vertical lines labelled `CENTRAL n% VIEW` around the unmasked centre.

Two frames captured during testing on 2026-09-12 with an earlier version: the overlay is that of v0.4.0's `annotate`, and the areas fall in v0.4.0's bands.



Passing a red pillar: state PASS\_RED\_RIGHT, steer +17.3 at heading 100.0; TF-Luna L 7, C 175, R 223 cm; track 18, red, confidence 0.92, 5 of 5 hits, bottom 0.63, area 0.0076, band close; orange tape boxes ahead and the red target column drawn.



A run that ended in FAULT: a red pillar 5 cm from the centre TF-Luna (L 99, C 5, R 56 cm), track 1 at confidence 0.86, bottom 1.00, area 0.0232, band too close; both tape lines are visible and the magenta corner reference is drawn. The fault reason is printed to the terminal, not to the view, and was not captured.

## 10. What the 2026-09-18 main.py changes

The file now in the repository has 6,663 lines against 4,771 in v0.4.0: about 2,430 lines added and 540 removed. This section lists the changes from the code; where the code gives a reason in a comment or docstring, it is quoted or paraphrased. The team removed the side rollers after this work ([vehicle §7](#)).

**Walls.** Three additions keep the car in the middle of its lane:

- PILLAR\_RECENTER follows every pillar pass. It counter-steers 22° against the pass for 0.18 to 0.45 s and moves on once the heading has swung back past straight, "proof it actually cancelled the drift, not just a timer"; it then

settles onto the lane centre from the heading and both side TF-Lunas (both walls 8 to 110 cm away, heading within 10°, gain 0.5, at most 18 of steer, 3 confirmations).

- After a corner the same kind of side-distance correction recentres the car (`ROUND2_RECENTER_*`: gain 0.45, at most 14, heading within 8°).
- `CORNER_CLEARANCE_RECOVERY` reverses until the front has 30 cm of clearance and re-checks the corner while stopped; it ends in `FAULT` if both side walls are too close for a corner reverse, or if the ESP32 does not report the reverse as applied. `TURN_SIDE_CLEARANCE_CM` is 20 (501 in v0.4.0, which carried 20 as a comment).

**Corners.** The corner approach is entered after three fresh front readings of 105 cm or less ("Enter the real corner approach after three fresh geometry samples", `CORNER_ZONE_LOCK_*`); after a corner the recognisers stay disarmed until the centre sensor sees a clearly open straight (130 cm, 3 samples), while the camera keeps detecting pillars. New states: `PRE_TURN_BACKUP` (0.28 s back at speed 88 before a turn) and `POST_TURN_SCAN_HOLD`, a 0.2 s stop after the post-turn reverse "so several stationary frames can prove a nearby pillar". The post-turn reverse is 0.64 s at speed 125 (2.8 s in v0.4.0).

**Pillars.** Two per straight, the central view and the parking-lot rule (§9). The red target column moved from 25 % to 38 % of the width, the near band starts at 3,200 px×px (4,000), the pass speed is 80 (100) and the release speed 80 (50).

**Parking at the end.** `ENABLE_PARKING_IN = True`. After the twelfth corner: `PARKING_ENTRY_SEARCH` (the final straight, pillars still handled), `PARKING_BLOCK_SEARCH` (the camera looks for the limitation), `PARKING_IN_ENTER` (camera-guided entry), `PARKING_IN_STOP_HOLD` (0.2 s), `PARKING_IN_REVERSE_LEFT` (3.0 s at full left lock), `PARKING_IN_REVERSE_RIGHT` (0.2 s), then `COMPLETE`. v0.4.0's `PARKING_IN_ANTI_*` and `PARKING_IN_CLOCK_ARC` states are gone. The team plans to park with `parallel_parking.py` at APAC instead (§8).

**Run log.** `RunCsvLogger` writes every run to `run_logs/round2_run_<date_time>.csv` (§7).

#### Values in §1-§6 that no longer hold:

| §       | v0.4.0                                                                                        | 2026-09-18                                                  |
|---------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1       | drive speed 210                                                                               | 215                                                         |
| 1, 5, 6 | <code>COMPLETE</code> after the twelfth corner; parking at the end off                        | parking-in follows the twelfth corner                       |
| 3, 5, 6 | last-resort front trigger 10 cm ( <code>PYTHON_EMERGENCY_RELEASE_CM</code> )                  | 5 cm, deeper inside the TF-Luna's specified 0.2 m minimum   |
| 4       | post-turn reverse 2.8 s                                                                       | 0.64 s at speed 125, after a 0.28 s reverse before the turn |
| 5       | same pillar ignored for 0.6 s after a pass ( <code>PILLAR_REACQUIRE_COOLDOWN_SECONDS</code> ) | removed; two pillars per straight instead                   |
| 3       | corner reverse at speed 70 for at most 0.8 s; deep recovery at speed 55                       | 200 for at most 0.5 s; 75                                   |
| 2       | obstacle corner confirmed on 5 samples in 0.30 s within 8°                                    | 3 samples in 0.25 s within 12°                              |

## 05 Engineering decisions

*Each change as a decision: what it replaced, why, the evidence, and the cost (criterion 4).*

### 7. Why each change was made

Each change as a decision: what it replaced, the reason the team gives, the evidence in this repository, and what it costs. The largest test-driven change in this vehicle's line is in the Nationals record: the wheel-shed failure of 2026-08-06 led to the space-frame redesign that every later frame descends from ([1 - Mobility §3, D8](#)).

| Change                                                           | Instead of                                       | Reason (team)                                                                                                                                                                                | Evidence here                                                                                                                                                                                                                                                                 | Trade-off                                                                                                                      |
|------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| New 84-step frame, 332 g bare                                    | the Nationals 69-step space-frame                | a sturdier structure                                                                                                                                                                         | build file; the car weighs 862 g against 785 g at Nationals (measured, all changes together)                                                                                                                                                                                  | 77 g more to accelerate and stop; stiffness not measured                                                                       |
| Steering column turning the whole front axle, 60° each way       | single-servo Ackermann                           | a smaller turning circle                                                                                                                                                                     | turning circle not measured                                                                                                                                                                                                                                                   | the front wheels no longer follow Ackermann geometry, so they scrub in tight turns                                             |
| Wheels: front 45 mm, rear 55 mm tyres                            | front 41 mm, rear 55.6 mm                        | rear tyres for grip                                                                                                                                                                          | not measured                                                                                                                                                                                                                                                                  | none recorded                                                                                                                  |
| Camera on a 24.8 cm tower, 12° down                              | a front arm at about 10 cm                       | a better view of the track                                                                                                                                                                   | measured mount, derived geometry: pillar bases at 0.5 m and 1.0 m are 12.5° apart against 5.6° before, a 2.2 × stronger range signal (§4)                                                                                                                                     | a 19 cm blind zone ahead of the wheels and a view over the walls (§4); a taller top, its centre-of-gravity height not measured |
| Six WS2812 LEDs on the front frame                               | ambient light                                    | fewer colour mistakes                                                                                                                                                                        | 0.07 A at the set brightness by the rule of thumb; Buck-2 idle 0.13 A measured with them (§5); misread counts not measured                                                                                                                                                    | data at 3.3 V may flicker (§6)                                                                                                 |
| Side rollers fitted, then removed                                | none                                             | fitted to keep the car moving on wall contact; removed once the rewritten wall-avoidance code made them unnecessary, and because they blocked the Raspberry Pi's ports (reported 2026-09-22) | the team ran the car in both configurations without trouble (team; the number of runs is not recorded); the 2026-09-18 <code>main.py</code> adds lane recentering after every pillar pass and after corners, and a corner-clearance recovery ( <a href="#">software §10</a> ) | without rollers a wall contact stops the car; only the Pi's collision escape (§6) remains                                      |
| Firmware front stop removed (2026-09-10)                         | a stop under 18 cm on the front TF-Luna          | stopping the drive motor ended a competition run                                                                                                                                             | one run ended (team)                                                                                                                                                                                                                                                          | collisions are handled by the Pi's distance logic alone (§6)                                                                   |
| Parallel parking at the end ( <code>parallel_parking.py</code> ) | direct entry ( <code>partial_parking.py</code> ) | 15 points for a full parallel park against 7 (2026 rules, scoring element 1.8.2)                                                                                                             | rules; no parking attempt logged yet ( <a href="#">software §8</a> )                                                                                                                                                                                                          | a longer manoeuvre beside the limitations; touching one ends the round (rule 9.24.7)                                           |

**The 2026-09-10 front-stop decision.** The firmware used to stop the drive motor when the front TF-Luna read under 18 cm (released at 24 cm) or when its reading went stale, and that stop ended a run. On 2026-09-10 `forwardHardStopIsActive()` was changed to return `false`, with the comment "Disabled: stopping the drive motor here ends a competition run". Collisions are now handled on the Pi: back away at speed 80 for 0.30 s, require at least 3 cm of progress, at most 2 attempts, release at 30 cm. The team kept the earlier versions as backups, which are not committed: `obstacle-challenge-final-code_withLED.ino.before-remove-front-brake-20260910.bak` (MD5 `a0f20dd94d2dce1a27c2fc72e3700922`) and `obstacle-challenge-final-code_withLED.ino.before-parking-override-`

20260910.bak (MD5 c93dd67690d2153980442369aa3df41d). The same day added a PARKING\_OVERRIDE\_ON serial command to the firmware; main.py never sends it, so it has no effect in a run.

## 06 Reproducibility

*How to rebuild and run the software exactly as committed (criterion 5).*

The Obstacle Challenge code the vehicle runs at WRO Future Engineers APAC 2026. The vehicle is described in [docs/apac\\_2026\\_vehicle.md](#), and how `main.py` drives the run, with its state diagrams, in [docs/apac\\_2026\\_software.md](#). The two parking modules are described in its [section 8](#).

| File                                                                                                     | Runs on        | What it does                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">obstacle-challenge/main.py</a>                                                               | Raspberry Pi 5 | Camera, pillar and line detection, and the driving decisions; sends drive commands to the ESP32                                                                                                                                                                                     |
| <a href="#">obstacle-challenge/parallel_parking.py</a>                                                   | Raspberry Pi 5 | Parallel parking at the end of the run ( <code>BUILD_ID = "slot-guided-entry-v29"</code> ): finds the two parking-lot limitations with the camera, drives into the gap and straightens up. Runs on its own and imports <code>main.py</code> ; <code>main.py</code> does not call it |
| <a href="#">obstacle-challenge/partial_parking.py</a>                                                    | Raspberry Pi 5 | Direct entry into the parking space on a locked heading, reusing the parallel module's approach and block search. Runs on its own; <code>main.py</code> does not call it                                                                                                            |
| <a href="#">obstacle-challenge/esp32/obstacle-challenge-final-code/obstacle-challenge-final-code.ino</a> | ESP32          | Drive motor, steering servo, LEDs, the three TF-Luna sensors and the BNO055 through the PCA9548A; sends telemetry to the Pi                                                                                                                                                         |
| <a href="#">tools/lab-calibration.py</a>                                                                 | Pi or laptop   | Interactive Lab colour-threshold picker (trackbars), with the same image preprocessing as <code>main.py</code> ; how it is used at a venue is in <a href="#">vehicle §4</a>                                                                                                         |
| <a href="#">tools/camera-settings.py</a>                                                                 | Pi or laptop   | Interactive brightness, contrast and gamma preview                                                                                                                                                                                                                                  |

### Serial link

115200 baud. `main.py` opens `/dev/ttyUSB0` on the Pi (`COM3` is its Windows bench setting).

- Pi to ESP32: `speed,direction,steer` and a newline. `speed` is the 8-bit PWM duty (0 to 255); `steer` is a correction of -60 to +60 around the servo centre of 85.
- ESP32 to Pi: telemetry lines `T,seq,ms,heading,left,center,right,validMask,hardStop,appliedSpeed`.

### Settings as committed

| Setting                             | Value                                                                                                                                                                                                                                                                                                    | File                                                                  |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Drive speed                         | <code>SPEED = 215</code> (84 % duty; 210 in v0.4.0)                                                                                                                                                                                                                                                      | <code>main.py</code>                                                  |
| Turning, parking, collision escape  | <code>100, 70, 80</code>                                                                                                                                                                                                                                                                                 | <code>main.py</code>                                                  |
| Leave the parking area at the start | <code>ENABLE_PARKING_EXIT = True</code>                                                                                                                                                                                                                                                                  | <code>main.py</code>                                                  |
| Park at the end                     | <code>ENABLE_PARKING_IN = True</code> : <code>main.py</code> parks with its own routine ( <a href="#">software §10</a> ). At APAC the team plans to park with <code>parallel_parking.py</code> ( <a href="#">software §8</a> ); the <code>main.py</code> that hands over to it is not yet in this folder | <code>main.py</code>                                                  |
| Run log                             | <code>ENABLE_RUN_CSV_LOGGING = True</code> : every run writes <code>run_logs/round2_run_&lt;date_time&gt;.csv</code> beside <code>main.py</code> , one row per telemetry sample and one per event                                                                                                        | <code>main.py</code>                                                  |
| Parking modules                     | run on their own, e.g. <code>python parallel_parking.py --direction clockwise --last-pillar auto</code>                                                                                                                                                                                                  | <code>parallel_parking.py</code> ,<br><code>partial_parking.py</code> |

| Setting                                 | Value                                                                                               | File                |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------|
| Parking run log (CSV and video per run) | ENABLE_RUN_LOGGING = False                                                                          | parallel_parking.py |
| Servo centre                            | SERVO_CENTER = 85                                                                                   | firmware            |
| Mux channels                            | TF-Luna left 0, centre 3, right 2; BNO055 4                                                         | firmware            |
| Front hard stop                         | disabled since 2026-09-10 (forwardHardStopIsActive() returns false); see <a href="#">vehicle §7</a> | firmware            |

Both parking modules need `parking_camera_calibration.json` beside them. It depends on how the camera is mounted, so it is made on the vehicle with `python parallel_parking.py --calibrate` and is not in the repository.

On the Pi, `main.py` sets GPIO3 high once it has initialised and is waiting for the start signal, and low when it exits.

## Provenance and checksums

Copied on 2026-09-21 from the team coach's working repository (private; commit `63c2b31`). The two parking modules arrived on 2026-09-23 as a Google Drive download (`drive-download-20260922T184733Z-1-001.zip`, which dates them 2026-09-22 03:49 and 04:26) and are stored unchanged. The ESP32 file was `obstacle-challenge-final-code_withLED.ino`; it is renamed here and placed in a folder of the same name, as the Arduino IDE expects. The files are stored byte-for-byte (see [.gitattributes](#)), so these MD5s can be checked on any checkout:

| File                                           | MD5                              |
|------------------------------------------------|----------------------------------|
| <code>main.py</code>                           | 9b7eb750d26e9df2448196452a0b5cbe |
| <code>obstacle-challenge-final-code.ino</code> | 28a6cdfba6ad9164471c5e9b56110a0c |
| <code>lab-calibration.py</code>                | ad3b6f191bb384d708ae5894bbf4d70d |
| <code>camera-settings.py</code>                | f02e7f9eaae329e8867e965ac4f98511 |
| <code>parallel_parking.py</code>               | 6e8112e6c355421a43c53fd91433d434 |
| <code>partial_parking.py</code>                | 7e4ef1645d1579f5c2f99f6063576c67 |

`main.py` was replaced on 2026-09-23 by the version dated 2026-09-18, from a second Google Drive download (`drive-download-20260922T190246Z-1-001.zip`) whose `cmd.txt` shows the coach starting `final-code/obstacle-challenge-final-code/main.py`. The version it replaces (MD5 `238dd1af810112d09598e433b483d5f8`), which [software §1-§6](#) describes, stays in the history ([the file before the change](#)). The ESP32 firmware in the September downloads is byte-identical to the committed file (MD5 `28a6cdfb`). Every `main.py` name the parking modules use also exists in the 2026-09-18 file.

Not included, because none of them runs on this vehicle: the build without LEDs, an older firmware for a smaller robot, backup copies (`.bak`), component tests and the rest of the coaching repository. The Open Challenge code is not in this folder.

Also left out from the September downloads: earlier and experimental versions of `main.py` (`main_16_Sep.py`, `mainversion2.py`, `corner_slot_filler.py`) and standalone tests (`parking_entry_only.py`, `cornering_sequence.py`).

# Part II

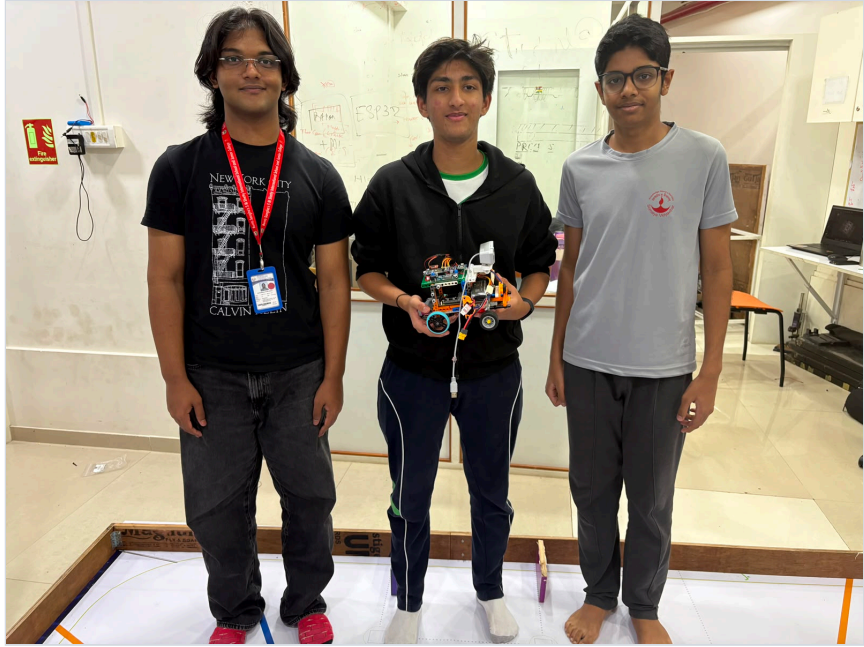
India Nationals journal, rev 3 (24 August 2026)

The first vehicle, kept unchanged. Part I describes the car that replaced it.

WRO FUTURE ENGINEERS 2026

# FINAL ENGINEERING JOURNAL

TEAM TED DRIVE



*Formal team photograph with the final competition vehicle*

|                   |                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------|
| Team members      | Ethan Fernandes — <b>Software</b> · Tejas Sirikonda — <b>Mechanical</b> · Diaan — <b>Electronics</b> |
| Coach             | Amey Chavan (OMOTEC)                                                                                 |
| Competition       | WRO Future Engineers 2026 · India Nationals, Hyderabad                                               |
| Competition dates | 26–28 August 2026                                                                                    |

**Repository:** [github.com/teddriveomo/wro-2026-future-engineers](https://github.com/teddriveomo/wro-2026-future-engineers)

Prepared 24 August 2026 (rev 3) against WRO 2026 General Rules, Appendix C (Engineering Journal evaluation).

# 01 Project Snapshot

*A concise map of the final system, evidence and current validation state*

Our vehicle is a fully autonomous four-wheel car built around a rigid LEGO Technic hybrid frame, single-servo Ackermann steering and a rear-drive differential. An ESP32 performs the time-critical sensing and steering loop, while a Raspberry Pi 5 handles pillar perception. The architecture was intentionally split so that a slow or failed vision process cannot stall steering.

| Item               | Final / current state                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Final robot mass   | 785 g, including competition electronics and battery                                                                                                                                                                                                                                                                                                                                                                                       |
| Open Challenge     | Implemented and tested; latest reported three-lap result: 24 s                                                                                                                                                                                                                                                                                                                                                                             |
| Obstacle Challenge | Competition stack committed at <code>src/Round 2/competition/</code> : ESP32 firmware ( <code>test.cpp</code> — PID heading hold, 14-state machine, reverse-arc corners, PAUSE→PASS→SIDE→YAW-BACK pass sequence) + Pi detector ( <code>detector.py</code> — 2-class green/red ONNX, 5-of-7 vote, 7-column offset LUT). Earlier stacks (3-class detector line, CIELab picker) retained in-repo as evidence. Final timed track test pending. |
| Parking            | Parking-lot <b>start</b> / <b>exit</b> implemented in the competition firmware (boot S-curve: reverse → 45° swing → straight leg → align, heading-verified). End-of-run parallel park-in: shimmy controller <b>written but not yet run</b> (branch <code>feat/round2-park</code> ); magenta bay-marker detection exists in the 3-class detector line.                                                                                      |
| Main power switch  | Installed on final vehicle (with separate start button — WRO two-button start sequence)                                                                                                                                                                                                                                                                                                                                                    |
| Steering           | MG90 metal-gear servo; Ackermann; 64–136° travel about a 106° centre                                                                                                                                                                                                                                                                                                                                                                       |
| Range / heading    | 3× TF-Luna through PCA9548A + BNO055 absolute yaw                                                                                                                                                                                                                                                                                                                                                                                          |
| Compute            | ESP32 for motion; Raspberry Pi 5 for vision                                                                                                                                                                                                                                                                                                                                                                                                |
| Drive              | N20 DC motor → TB6612FNG → 20T:28T → LEGO differential                                                                                                                                                                                                                                                                                                                                                                                     |

**WRO documentation target.** Appendix C awards up to 6 points each (scored 0/2/4/6) for Mobility, Power/Sensors, Software/Obstacle Strategy, Systems Thinking, and Reproducibility/GitHub Quality. This journal is organised in exactly those five areas and emphasises decisions, tests, iterations and rebuildability rather than appearance alone.

| Section                                     | What the judges can find                                                                                                                              |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Mobility & Mechanical Design             | Chassis, drivetrain, steering geometry, torque/traction/speed derivation, CAD of printed parts, mechanical iteration, final mass                      |
| 2. Power & Sensor Architecture              | Power tree, sensor selection, placement, I2C architecture, calibration and failure handling                                                           |
| 3. Software & Obstacle Strategy             | Control-design rationale (P vs PID), Open control flow, firmware logic excerpt, Round 2 state machine, perception pipeline with code, tuning evidence |
| 4. Systems Thinking & Engineering Decisions | Constraints, trade-offs, rejected alternatives, iteration cycles and risk mitigation                                                                  |
| 5. Reproducibility & GitHub Quality         | Repository structure, code/CAD/wiring/video evidence, environment and rebuild workflow                                                                |
| Testing & Final Status                      | Latest Round 1 result, pending Round 2/parking, pre-competition verification items                                                                    |

## 02 Engineering Process

*The journal records how the design changed, not only what the robot looks like now*



*Informal engineering-session photograph: reviewing the field, pillar placement and robot behaviour during development*

The project progressed through repeated plan–build–test–improve cycles. Several ideas that looked correct on paper were changed after measured failures. We deliberately kept those reversals in the repository because they explain why the final system exists in its present form.

### Engineering iteration timeline

| Date      | Milestone                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------|
| 26 Jun    | Initial motion firmware — heading hold + corner detection                                                   |
| 19 Jul    | HSV pipeline measured; failure modes quantified                                                             |
| 26 Jul    | Clean dataset split rebuilt after leakage audit; NanoDet line adopted                                       |
| 5 Aug     | Field stack reworked; repo restructured to the five-criterion layout                                        |
| 9–10 Aug  | Chassis stiffness redesign validated; steering retune + correction-sign fix                                 |
| 11–12 Aug | CAD collected, videos recorded/uploaded, magenta class added, documentation snapshot                        |
| 19 Aug    | 3-class YOLO26n v2 retrained on the leakage-fixed split (Colab, 80 epochs)                                  |
| 21 Aug    | Round-1-lineage obstacle firmware + Pi vision sender committed and merged to main                           |
| 24 Aug    | Competition Round-2 stack committed (src/Round 2/competition/); capture tooling merged; final journal rev 3 |

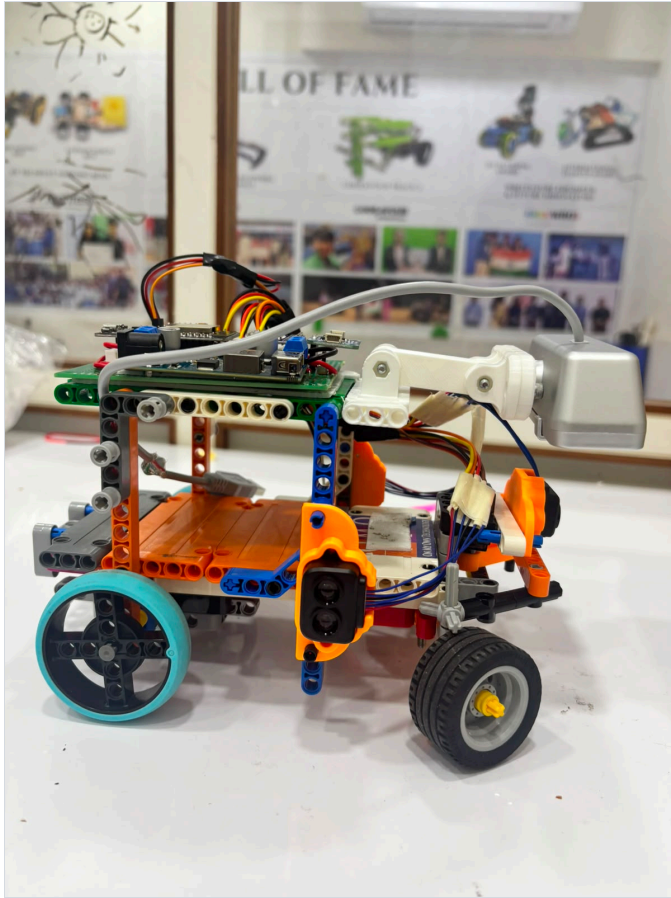
### The strongest iteration examples

- Corner detection changed from a level test to a rising-edge event because a level condition repeatedly stepped the target heading and could make the robot spin.
- The first LEGO frame flexed under drivetrain load and shed wheels. Instead of immediately replacing it with a printed chassis, we redesigned the LEGO space-frame to be much stiffer; later sustained runs no longer lost wheels.

- The first neural pillar detector performed well on the clean validation split but transferred poorly to real field footage. We fell back to a per-venue calibrated CIELab colour picker, retrained a 3-class YOLO26n on a leakage-fixed split, and finally converged on the 2-class grid/pause-pass competition stack — **every stage is kept in the repository as evidence** (§08–§09).
- The steering correction **sign was inverted** until the 10 Aug retune — the servo initially fought the heading error instead of nulling it. Caught on the mat, fixed in one commit, and documented rather than hidden (§07).
- Sensor power routing was reworked so the MG90 servo receives 5 V from the supply rail while the ESP32 provides only the control signal, reducing a previously identified failure path.
- A dedicated main power switch has now been installed on the final vehicle, completing the WRO start sequence alongside the separate start button.

### 03 Criterion 1 — Mobility & Mechanical Design

*How the chassis, drive and steering were chosen, measured and refined*



Side view: electronics stack, rear drive wheel and front steering



Opposite side: camera mount, LiDAR and drivetrain packaging

#### Final mechanical architecture

| Subsystem          | Final design                                                                  | Engineering reason                                                                                                 |
|--------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Chassis            | Rigid LEGO Technic hybrid space-frame with printed mounts                     | Rebuildable, light and sufficiently rigid after the flex-induced wheel-loss failure was diagnosed.                 |
| Steering           | One front steered axle, single MG90 metal-gear servo, Ackermann linkage       | Complies with the non-skid steering requirement and gives a predictable geometric steering response.               |
| Drive              | Rear axle driven by one N20 motor through TB6612FNG and LEGO differential     | Both driven wheels remain mechanically connected; the differential allows wheel-speed difference during cornering. |
| External reduction | 20-tooth pinion → 28-tooth crown = 5:7 (1.4:1), counted on the vehicle 11 Aug | Trades some speed for torque/control without needing a drivetrain rebuild.                                         |
| Wheels             | Rear 55.6×14 mm; front 41×21 mm                                               | Larger driven rear wheels improve distance per revolution; smaller front wheels suit the steering geometry.        |
| Mass               | 785 g final                                                                   | Only 52.3% of the 1.5 kg limit, leaving 715 g of margin.                                                           |

Printed parts — CAD (designed in Fusion 360, exported STL, in models/)



N20 motor clamp — LEGO-pitch holes      Servo horn beam — Ackermann link      TF-Luna enclosure — LEGO-pitch

Each part solves a LEGO↔commodity-hardware interface: the clamp holds the round N20 body at LEGO pitch, the horn beam converts the servo horn into a Technic-pin link, and the enclosure fixes each TF-Luna's aim height and yaw. The LEGO chassis itself is captured as chassis\_lego\_current.lxfml plus step-by-step build instructions (PDF) in models/chassis/.

Torque, traction and acceleration (derived; datasheet-class motor figures, labelled accordingly)

Speed answers "how fast can it go" — this answers "can it get there, and what stops it first." The motor is a GA12-N20-600 class unit: 12 V rated, 600 rpm no-load, 1:50 gearbox, 0.18 kg·cm rated / 0.65 kg·cm stall torque. The external stage multiplies motor torque by  $R_{ext} = 1.4$  (20T→28T, counted on the vehicle):

|       | Motor (datasheet-class) | × 1.4 → wheel | Tractive force at r = 27.8 mm |
|-------|-------------------------|---------------|-------------------------------|
| Rated | 0.18 kg·cm = 0.0177 N·m | 0.0247 N·m    | <b>0.89 N</b>                 |
| Stall | 0.65 kg·cm = 0.0637 N·m | 0.0892 N·m    | <b>3.21 N</b>                 |

**What stops the vehicle first — and it is genuinely close.** The traction limit is  $\mu \cdot N_{rear}$ . At the measured 785 g with 50–60% of weight on the driven axle and rubber-on-painted-mat  $\mu \approx 0.5\text{--}0.9$ , the traction ceiling spans roughly **1.9–4.2 N**. Against a stall tractive force of **3.21 N** the design sits *on the boundary*: at the slick end the wheels break traction before the motor stalls; at the grippy end the motor stalls first. The distinction is not academic — it decides whether a wall strike stalls the motor (current spike, TB6612 thermal event) or spins the wheels (no spike, but no recovery either). A front/rear axle-load measurement settles the regime and is listed as an open verification (§13).

**Acceleration.** Motor torque falls roughly linearly from stall toward rated as speed rises; a mid-ramp average of ~2 N on 0.785 kg gives  $\approx 2.5\text{ m/s}^2$ , so the vehicle reaches its derived 1.15 m/s ceiling in well under a second and under half a corridor width — it is at speed before the first corner, which is why the final run-on behaves as a distance rather than a ramp.

**Does the ratio give enough speed?** The driving corridor is ~1000 mm inside a 3200 mm mat, so a centre-line lap is roughly  $4 \times 2.2\text{ m} \approx 8.8\text{ m}$  and three laps  $\approx 26\text{ m}$  — about 23 s of pure driving at the 1.15 m/s ceiling before corner slowdowns. The 5:7 stage is therefore not the binding constraint on round time, which is the justification for leaving it rather than re-cutting for speed.

## 04 Steering Control Design & Mechanical Testing

*Why the controller is proportional-only, and the test evidence that changed the physical design*

### Control design — Round 1: PID evaluated, P-only selected

The steering loop converts wrapped heading error ( $\pm 180^\circ$ ) into a servo correction. We evaluated the full PID family against the specific structure of this task before choosing the final law:

| Term     | What it would do here                                                                  | Decision                                                                                                                                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P</b> | Correction proportional to heading error — restoring torque toward the target heading. | <b>Kept.</b> Gain = 1.0 servo-degree per degree of error; a $90^\circ$ corner command drives the servo straight to full lock, so corners need no separate manoeuvre.                                                                                                                                                                            |
| <b>I</b> | Accumulates error over time to remove steady-state offset.                             | <b>Rejected.</b> Our target heading is <i>deliberately stepped</i> by $90^\circ$ at each corner — a commanded discontinuity. An integrator winds up across the entire transit of every corner and then discharges as overshoot on the next straight. The plant also has no steady-state error source that P + geometry does not already handle. |
| <b>D</b> | Damps by reacting to error rate.                                                       | <b>Rejected.</b> The BNO055 yaw is quantised; differentiating it amplifies quantisation noise into servo chatter, wearing the MG90 and adding jitter on straights.                                                                                                                                                                              |

The final law is **P = 1.0 deg/deg with a 2.0° deadband**, output clamped to the measured mechanical window  $64\text{--}136^\circ$  about the  $106^\circ$  centre (asymmetric  $-42^\circ/+30^\circ$ ). The deadband suppresses hunting inside the sensor's noise floor; the clamp guarantees no software state can command the linkage past a hard stop. Straight-line gain can then be tuned for stability rather than corner aggressiveness, because corners are executed by the target step saturating the P term — not by a separate turn routine.

**Round 2 runs the full PID — and the difference is structural, not taste.** The obstacle firmware (src/Round 2/competition/test.cpp) holds heading with **Kp = 1.5, Ki = 0.02, Kd = 0.15** on the wrapped  $\pm 180^\circ$  error: P and I are gated by the heading deadband, the integral is hard-clamped at  $\pm 50$  (anti-windup), and the total correction is clamped to  $\pm 20$  servo-degrees. Round 2's disturbance profile is different from Round 1's — the pass sequence holds sustained steering offsets and re-join arcs rather than stepping the target, so a derivative term is justified damping the return to the lane, and a small, clamped integral trims residual bias during long offset holds. Both clamps bound each term's failure mode, which is exactly what makes I and D admissible here after being rejected for Round 1. Each round runs the controller its disturbance profile justifies.

**Iteration evidence — the sign bug.** Until the 10 Aug retune the correction *sign was inverted*: the servo steered into the error instead of nulling it, and the "centre" and limits were also stale ( $45\text{--}135^\circ$  about  $90^\circ$  from an earlier linkage). The on-mat symptom — divergence instead of convergence — was traced in one session; the fix (sign flip, centre  $106^\circ$ , limits  $64\text{--}136^\circ$ , gain 1.0, deadband  $2.0^\circ$ ) is a single documented commit. We kept the wrong version in history because it is the reason the current numbers exist.

### Failure → diagnosis → redesign (chassis)

| Stage                   | Observed evidence                                  | Decision / result                                                             |
|-------------------------|----------------------------------------------------|-------------------------------------------------------------------------------|
| Initial sustained drive | Wheels worked loose from the axles within seconds. | Treat wheel retention as a frame/stiffness problem, not only a wheel problem. |

|                     |                                                                                                  |                                                                                                                       |
|---------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| First proposed fix  | A fully 3D-printed chassis was considered to add rigid hubs and retention.                       | Kept as a fallback, but not committed as the final answer.                                                            |
| Final fix           | Ground-up redesign of the LEGO space-frame; tighter packaging and a lower, more rigid load path. | Two later runs of 28 s and 29 s continuous driving recorded no wheel loss; the printed chassis rebuild was cancelled. |
| Final configuration | Same stiff frame used for all later work; full electronics and camera mounted.                   | Final competition mass measured at 785 g.                                                                             |

## Latest Open Challenge validation

| Test                     | Result                                             | Interpretation                                                                                             |
|--------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Three-lap Open Challenge | 24 s total                                         | 8 s per lap average; confirms complete three-lap operation on the latest reported test.                    |
| Automatic stop logic     | 12 corners observed, then final run-on and braking | Software counts 90° heading steps and stops after the third lap rather than relying on elapsed time alone. |
| Mass check               | 785 g                                              | Large legal margin below the 1.5 kg limit and less load on steering / drivetrain.                          |

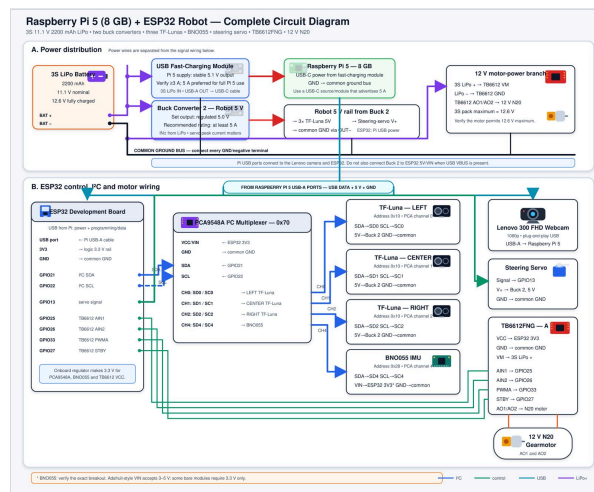
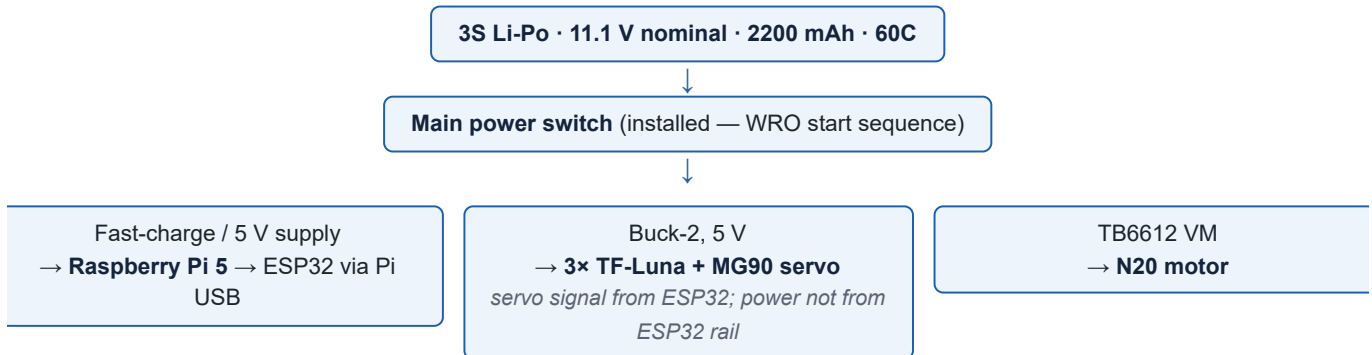
**Cross-checking the 24 s result against the drivetrain model.** 24 s over three laps means the average speed equals the driven lap length  $\div$  8 s. On the centre-line estimate ( $\sim 8.8$  m/lap) that would imply  $\sim 1.1$  m/s — implausibly close to the 1.15 m/s *unloaded* ceiling — so the result is consistent only with the shorter inner line the controller actually drives after each corner ( $\sim 6.5$ – $7.5$  m/lap), giving a loaded average of **0.8–0.95 m/s  $\approx$  70–83% of the no-load ceiling**. That is aggressive but physically coherent; a tape measurement of the driven line is listed as an outstanding verification (§13), and the loaded straight-line speed itself is re-measured by tape + timer before the rounds. This journal does not carry a straight-line speed number until that measurement exists.

The Open Challenge test is the strongest complete end-to-end validation currently available. The Obstacle Challenge and parallel-parking validation are kept separate in later sections because they are not yet complete; this journal does not convert unfinished work into claimed results.

## 05 Criterion 2 — Power & Sensor Architecture

*How sensing and power are organised so that the system remains reliable and reproducible*

### As-built power architecture



*Complete as-built circuit, photographed 11 Aug; drawn schematics (signal + power, v0.2) are versioned in schemes/*

### Power distribution decisions

- 3S Li-Po pack: 11.1 V nominal, 2200 mAh, 60C.
- Main power switch is installed in the final vehicle so the robot can be placed on the field switched OFF, then powered ON before the separate start command.
- The Raspberry Pi 5 is powered through the vehicle's regulated 5 V supply; the ESP32 is supplied from the Pi USB path.
- A separate 5 V buck rail powers the three TF-Lunas and MG90 servo. The servo control signal comes from the ESP32, but the servo is not powered from the ESP32 rail.
- The motor rail feeds TB6612FNG VM and the N20 separately from the 5 V electronics branch, reducing interaction between motor load and sensing/control.
- **Predicted power budget** (derived, versioned in docs/2\_power\_predicted\_budget.md): cruise draw  $\approx$  0.95–1.5 A at the pack  $\rightarrow$  23–37 three-minute runs per charge on 1760 mAh usable. Perception (Pi + camera) costs  $\approx$  3 $\times$  propulsion; the binding constraint is Pi 5 thermal throttling (82 °C onset), not pack capacity.

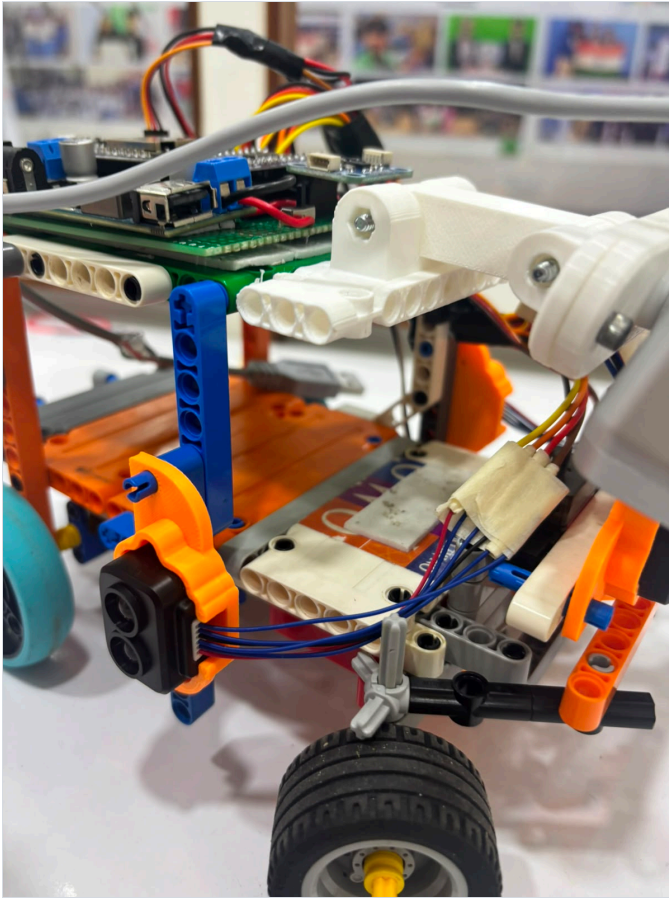
### Sensor map and why each device exists

| Device     | Qty / bus                           | Role                   | Reason for selection                                                                                               |
|------------|-------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------|
| BNO055 IMU | 1 · I <sup>2</sup> C 0x28 · mux ch4 | Absolute yaw / heading | Avoids three-lap drift associated with integrating gyro rate; heading targets are stepped by exact 90° increments. |

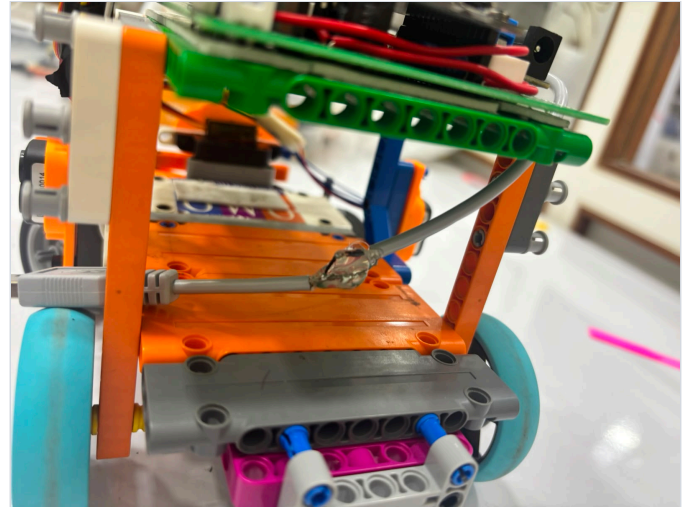
|               |                                            |                                           |                                                                                               |
|---------------|--------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------|
| TF-Luna LiDAR | 3 · I <sup>2</sup> C 0x10 · mux<br>cho/1/2 | Left, centre, right distance              | Narrow ~2° beam gives a sharper wall-<br>opening edge than the much wider ultrasonic<br>cone. |
| PCA9548A      | 1 · I <sup>2</sup> C 0x70                  | I <sup>2</sup> C channel multiplexer      | All TF-Lunas share address 0x10, so each<br>must be isolated on its own channel.              |
| USB camera    | 1 · Raspberry Pi 5                         | Red / green / magenta<br>visual detection | Colour + bearing for obstacles and parking<br>markers.                                        |

## 06 Sensor Placement, Calibration & Failure Handling

*The sensing layout was designed around field geometry and known failure modes*



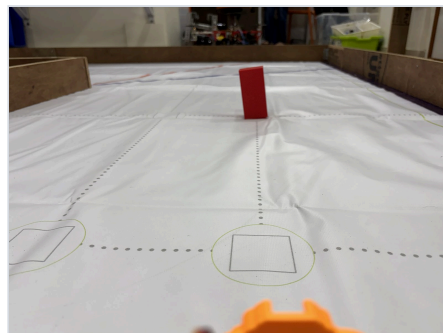
*Front sensor + camera mount as built — TF-Luna aim height and camera pitch fixed by the printed mounts (\$03 CAD)*



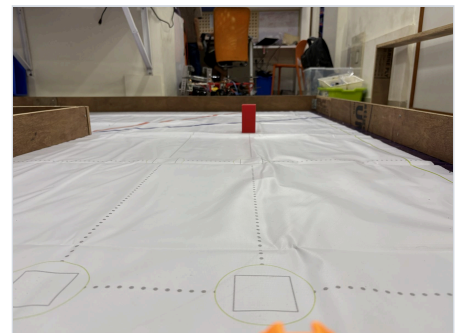
*Chassis midsection: mux, wiring loom and load path*



*Camera POV — near pillar*



*Camera POV — mid range*



*Camera POV — far; sets the minimum detectable box height*

### Geometry and bus architecture

The side TF-Lunas watch for the track wall to disappear. A corner is not treated as "distance is large"; it is treated as the transition from wall-present to an opening over 150 cm. That distinction makes sensor placement and beam width more important than absolute range accuracy. The centre TF-Luna provides forward clearance; in Round 2 it is the corner trigger (reverse-arc at 20 cm).

All three TF-Lunas respond at I<sup>2</sup>C address 0x10, so direct parallel connection would cause an address collision. The PCA9548A solves this by selecting one channel at a time: left on channel 0, centre on 1, right on 2, and the BNO055 on channel 4. The ESP32 I<sup>2</sup>C bus uses GPIO21 (SDA) and GPIO22 (SCL).

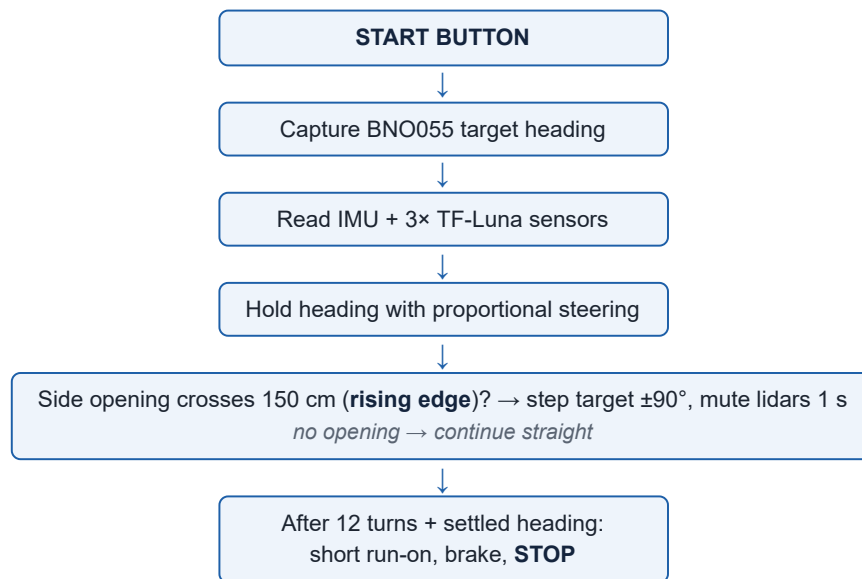
## Calibration and noise controls

- BNO055 heading is captured at the start button, after the robot has been positioned on the field, so the reference frame follows the actual starting orientation.
- Round 1 accepts a corner only on a rising edge with a 1 s lidar mute; Round 2's corner trigger is the front LiDAR at 20 cm with a 1 s cooldown — in both designs a pillar briefly occluding a side sensor cannot fire a turn.
- The competition detector's class map is **locked in code** ({0: green, 1: red}); the 3-class line's map ({0: green, 1: red, 2: magenta}) is likewise locked, because the exported ONNX carries no name metadata — the constant is always the map in use.
- Temporal voting — 5-of-7 agreeing frames with a 70 px spatial lock before a pillar is confirmed, and a 3-frame CLEAR debounce — prevents a single bad frame from commanding or cancelling a pass.
- In the 3-class detector line, the magenta parking-marker path includes a dedicated chroma floor and morphological opening to separate marker colour from reflection bleed on the mat.

**Failure-aware design.** The repository explicitly treats lighting changes, sensor occlusion, I<sup>2</sup>C address collisions, motor magnetic interference and perception stalls as engineering risks. The two-processor split ensures a perception stall does not block the steering loop.

## 07 Criterion 3 — Software Architecture & Open Challenge

*The motion controller is event-driven, heading-based and designed for repeatability*



### Open Challenge algorithm

- When the start button is pressed, capture the current BNO055 yaw as the target heading.
- Read the IMU and TF-Luna sensors continuously.
- Use proportional steering (\$04 control design) to reduce wrapped heading error while driving forward.
- Detect a corner only when a side distance crosses from wall-present to an opening greater than 150 cm.
- Step the target heading by +90° or −90° toward the opening; mute the lidars for 1 s so the opening cannot re-trigger while it remains visible.
- Count the completed 90° target-heading steps. After 12 turns, wait until heading is settled, perform the final short run-on, brake and stop.

**Firmware logic — verbatim excerpt from `src/Round 1/round 1/round 1.ino` (tunables + rising-edge corner detector)**

```

#define SERVO_MAX_ANGLE 136 // servo upper limit (deg)
#define SERVO_CENTER 106 // wheels-straight angle
#define STEER_GAIN 1 // servo deg per deg of heading error
#define OPENING_CM 150 // side reading above this = corner/opening
#define TURN_STEP_DEG 90 // heading change per detected opening
#define STEER_DEADBAND 2.0
#define LOOP_DELAY_MS 20
// ... inside loop(): heading always read, steering always live ...
bool lOpen = (l > OPENING_CM);
bool rOpen = (r > OPENING_CM);

bool turned = false;
if (lOpen && !leftWasOpen) { targetHeading -= TURN_STEP_DEG; turned = true; }
if (rOpen && !rightWasOpen) { targetHeading += TURN_STEP_DEG; turned = true; }

leftWasOpen = lOpen;
rightWasOpen = rOpen;

if (turned) lidarResumeAt = millis() + 1000; // corner taken -> mute lunas 1 s
// during the pause: no reads, no corner detection – but steering stays live
int servoAngle = steerToHeading(h, targetHeading);
  
```

## Why the rising-edge corner detector matters

A threshold condition such as "if distance >150 cm, turn" remains true for many control cycles while the sensor is looking into the open corner. Without memory of the previous state, the target heading could be stepped repeatedly by 90°, producing a spin. The rising-edge design fires once at the instant the wall disappears (`!Open && !leftWasOpen`), then the 1 s lidar mute plus the stored wall state require a genuine new edge before another corner can be detected. Steering remains live throughout the mute — the P-controller saturating against the stepped target is what physically executes the corner.

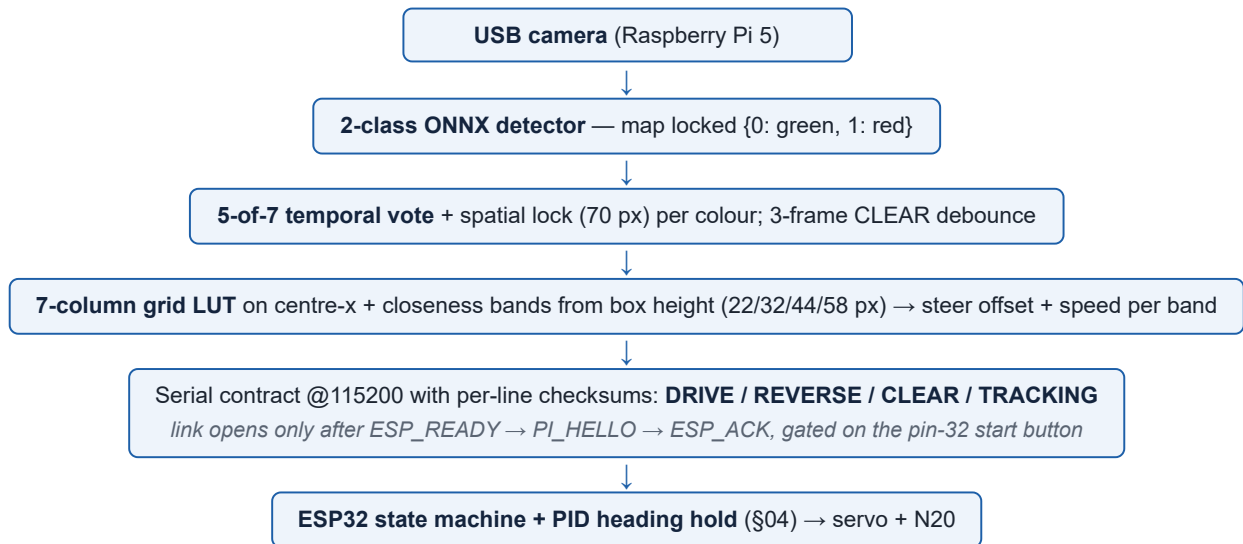
## Latest measured outcome

The latest team-reported Open Challenge test completed all three laps in 24 seconds (analysed against the drivetrain model in §04). It demonstrates the interaction of chassis rigidity, drivetrain, heading control, corner detection, lap counting and autonomous stop logic rather than testing any one module in isolation.

## o8 Obstacle Challenge Strategy

A staged pass sequence, a PID heading hold and a checksummed serial contract between the two processors

**Perception → decision → actuation pipeline (competition stack, src/Round 2/competition/)**



### Implemented state logic (ESP32, 14 states)

| State(s)                                                                             | Purpose                                            | Key behaviour / exit                                                                                                                                                         |
|--------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PARK_WAIT →<br>PARK_REVERSE →<br>PARK_PHASE1 →<br>PARK_STRAIGHT_LEG →<br>PARK_PHASE2 | Start inside the parking lot; S-curve exit at boot | Reverse out of the slot, 45° swing away, forward straight burst, counter-swing to align — heading-verified against the start reference, with an 8 s safety cap               |
| DRIVING_STRAIGHT                                                                     | PID heading hold and obstacle monitoring           | Confirmed pillar → pass sequence; front TF-Luna under 20 cm → corner                                                                                                         |
| PRE_TURN_REVERSE /<br>TURNING                                                        | Reverse-arc 90° corner                             | If a far cube is still in view, IMU reverse on the cardinal first (650 ms, to 28 cm clearance), then the reverse-arc turn; target snapped to the cardinal grid; 1 s cooldown |
| OBSTACLE_AVOIDING<br>(PAUSE → PASS)                                                  | Execute the pass on the commanded side             | 400 ms full stop at 45 px box height with pass steer held frozen, then wide pass (full 35° swerve floor) until the cube leaves the camera edge                               |
| PI_SIDE → PI_YAW_BACK                                                                | Re-join the lane after the cube is off-camera      | Drive straight until the matching side LiDAR (red = left, green = right) sees the cube at ≤20 cm — or a 1.6 s cap — then a fixed 28° yaw back toward the lane                |
| PI_REVERSE                                                                           | Recover when pillar geometry becomes too close     | Short 550 ms reverse bursts (max 3), then retry the pass — never latch                                                                                                       |
| ROBOT_STOPPED                                                                        | End after 12 counted turns (3 laps)                | Terminal state                                                                                                                                                               |

## Obstacle perception and pass-side decision

The Pi runs a 2-class green/red ONNX detector. Each colour keeps a 7-frame history: a pillar is *confirmed* only on 5-of-7 agreeing frames with a 70 px spatial lock, and a CLEAR must persist 3 frames — a single bad frame can neither command a pass nor cancel one. The nearest confirmed box is placed on a 7-column grid over centre-x with closeness bands from box height; the LUT emits a steering offset (up to the full 35° swerve for both colours — an earlier on-robot LUT that peaked below full lock let the body clip the cube, which is why the floor is now 35°) and a band speed. Red passes right, green passes left. At 45 px box height the sequence inserts a deliberate 400 ms full stop with the pass steer frozen, then commits to the wide pass; after the cube leaves the frame the firmware goes straight until the *matching side LiDAR* confirms the cube alongside, then yaws back a fixed 28°. A segment allows at most 2 passes before the next corner.

Pi-side detection & pass-side decision — verbatim excerpts from `src/Round 2/competition/detector.py`

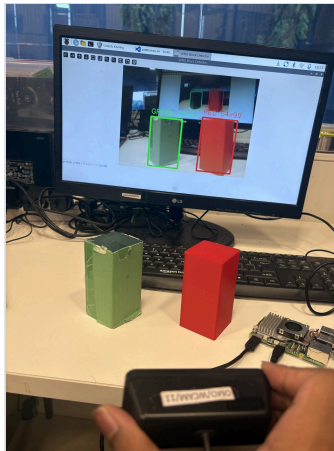
```
ONNX_MODEL_PATH = "best_ncnn.onnx"
CLASS_NAMES = {0: "green", 1: "red"}
VOTE_HISTORY = 7
MIN_VOTES = 5
CLEAR_HISTORY = 3
LOCK_MATCH_PX = 70
SWERVE_HEIGHT_PX = 45          # start the pass only when the cube is this tall
ALIGN_PAUSE_S = 0.40          # stop 400 ms at 45px, then swerve
GREEN_MIN_OFFSET = 35         # green pass-left needs full DIFF or the body clips
RED_MIN_OFFSET = 35           # red pass-right, same full swerve
# ... main loop: confirmed 5-of-7 per colour, nearest actionable box wins ...
cmd = (
    f"DRIVE,{primary_color},{cmd_info['speed']},{cmd_info['offset']},{",
    f"{cmd_info['phase']},{active_box['height']}"
)
if cmd != last_sent or due:
    serial_write(cmd)
elif decision == "CLEAR" and clear_counter >= CLEAR_HISTORY:
    serial_write("CLEAR")
```

**Model file:** the weights sit next to the script under the name in `ONNX_MODEL_PATH`; the 3-class v2 weights versioned in `../detector/models/` decode green/red at the same indices (0/1) if substituted — the extra magenta class is simply ignored by this map. **Lineage:** the 3-class detector line (with its leak-fixed per-class metrics, §09) and the venue-calibrated CIELab picker are retained in-repo as the measured evolution behind this stack.

**Current validation status.** The competition stack is committed and the components are individually exercised, but the final timed Round 2 track test has not yet been completed. This journal describes the implemented architecture and its component-level evidence without inventing a final race result.

## 09 Vision Iteration & Data-Driven Decisions

*The final perception method was chosen after multiple approaches were measured and rejected*



*Live detection on the Raspberry Pi 5 — dual-pillar frame with class + confidence overlay (bench setup in docs/img/)*

### Detector evolution

| Iteration                                  | Evidence                                                                                                                                                                                                                                                                                         | Engineering response                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| HSV / connected components                 | Very fast (~0.43 ms/frame), but failed with both missed pillars and false positives.                                                                                                                                                                                                             | Could not be rescued by a downstream verifier because missed proposals never reach the verifier.                              |
| tiny_pillar (111K params)                  | Lower wrong-side rate, but 9.7% no-call and lower overall decision performance.                                                                                                                                                                                                                  | Rejected in favour of stronger decision coverage.                                                                             |
| nanodet_lite (1.17M params)                | Clean-split macro F1 0.898 and pass-side accuracy 0.941; beat tiny model on six of seven metrics.                                                                                                                                                                                                | Selected as the neural winner on validation data.                                                                             |
| Field transfer check                       | Neural winner failed to transfer reliably to real field footage; repository had already documented the one-session-data limitation.                                                                                                                                                              | Per-venue calibrated CIELab picker adopted as the interim competition stack; failed neural path preserved as evidence.        |
| Magenta extension                          | Close-range reflection bleed joined marker colour into a large blob.                                                                                                                                                                                                                             | Dedicated magenta class, stronger chroma floor and morphological OPEN; parking-marker telemetry added.                        |
| <b>3-class YOLO26n-320 v2</b>              | Retrained 19 Aug on the leakage-fixed, group-wise split, 80 epochs: <b>mAP50-95 green 0.859 / red 0.792 / magenta 0.820.</b>                                                                                                                                                                     | Adopted for the 3-class detector line; weights (.onnx + .pt) and harness versioned in-repo.                                   |
| <b>Grid / pause-pass competition stack</b> | 2-class green/red ONNX with 5-of-7 vote + spatial lock; 7-column offset LUT with closeness bands; staged PAUSE→PASS→SIDE→YAW-BACK pass paired with the PID firmware. An earlier on-robot LUT peaked below full steering lock and let the body clip cubes — the offset floor is now the full 35°. | <b>Committed as the flashed Nationals configuration</b> (src/Round 2/competition/, §08); earlier stacks retained as evidence. |

### Validation integrity: the split was rebuilt after leakage was found

When validation numbers looked better than real behaviour, the team audited the dataset and found 25 duplicate image stems across the split boundary and 27 of 29 validation time buckets overlapping training. Instead of

adjusting the results, all previous numbers were withdrawn. A script then rebuilt a group-wise 473/124 split by capture bucket, and the models were retrained and re-measured. This was one of the most important systems-thinking decisions because it changed which evidence could be trusted — and it is why the v2 per-class numbers above are reported against the fixed split only.

### **Why venue calibration was adopted (and kept for the fallback)**

Measured colour centres shifted substantially between capture sessions: the repository reports a 24.0-unit movement in the fitted red Lab chroma centre between two sessions, larger than the normal tolerance budget. Condition-matched calibration therefore became a procedure: calibrate in the actual venue light during check time, save the calibration and reuse it headlessly for the run. This converts lighting from an uncontrolled assumption into a managed setup step — and the same discipline applies to the competition detector: verify it on the actual field in venue light before trusting bench numbers.

## 10 Criterion 4 — Systems Thinking & Engineering Decisions

*The robot was treated as one interacting system under constraints, not as separate parts*

### Primary constraints

| Constraint                             | Design response                                                                                                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 300 × 200 × 300 mm envelope            | Compact stacked electronics and narrow chassis; camera and sensor mounts integrated into the frame.                                                                                                          |
| ≤ 1.5 kg total mass                    | Final vehicle is 785 g, leaving 715 g margin for compliance and limiting drivetrain/steering load.                                                                                                           |
| Ackermann / non-skid steering          | One steered front axle with MG90 linkage; one mechanically connected rear drive axle through a differential (a differential gear on the driven axle is rule-legal; differential-drive steering is not used). |
| Full autonomy                          | Two-button procedure: main power switch followed by the dedicated start button; no manual control during a run.                                                                                              |
| No wireless during rounds (rule 11.10) | Pi radios disabled for every round; the competition stack opens no network paths — preview is a local window only and headless by default.                                                                   |
| Vision compute budget                  | Raspberry Pi 5 CPU-only; motion remains on ESP32 so perception load cannot block control.                                                                                                                    |
| Asymmetric scoring risk                | The pass-side system is tuned to avoid wrong-side decisions; an uncertain/no-call is treated differently from a confidently wrong call.                                                                      |
| Time before competition                | Engineering evidence was versioned continuously in GitHub so design changes and failures were not lost.                                                                                                      |

### Representative trade-offs

| Choice                               | Alternative                              | Why the final choice won                                                                                                                                                                                                                                                               |
|--------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rigid LEGO redesign                  | Full printed chassis rebuild             | Testing showed stiffness was the root issue; the lighter/cheaper LEGO redesign fixed wheel retention without a complete material change.                                                                                                                                               |
| BNO055 absolute heading              | MPU6050 gyro integration                 | Absolute/fused yaw avoids accumulating turn-rate drift over three laps.                                                                                                                                                                                                                |
| TF-Luna LiDAR                        | HC-SR04 ultrasonic                       | Narrow beam produces a sharp opening edge, improving corner event timing.                                                                                                                                                                                                              |
| Two processors                       | Single-board all-in-one compute          | Vision delays cannot pause heading hold and steering.                                                                                                                                                                                                                                  |
| P-only steering (R1) / full PID (R2) | One controller for both rounds           | Round 1's stepped target rejects I and D (§04); Round 2's sustained pass offsets and re-join arcs justify clamped I and D. Each round runs the law its disturbance profile earns.                                                                                                      |
| Staged grid-LUT pass strategy        | Single-threshold swerve / earlier stacks | The pass is decomposed into measurable stages (400 ms pause at 45 px, edge-locked pass at full 35°, LiDAR-gated SIDE, fixed 28° yaw-back), each with its own tunable — failures localise to a stage instead of one opaque gain. Earlier stacks kept in-repo as the measured evolution. |
| Keep 20T:28T external stage          | Regear for more speed                    | Lap arithmetic (§03) shows the ratio is not the binding constraint on round time; control and cornering matter more than an unloaded maximum.                                                                                                                                          |

**Engineering principle used throughout.** We chose X instead of Y when tests or constraints gave a reason. When an earlier decision was wrong — the steering sign, the leaky split, the first neural detector — the repository keeps the original reasoning and the reversal so that the engineering process remains auditable.

## 11 Risk Register & Iteration Cycles

*Known failure modes were documented with mitigations rather than hidden*

| Risk                                              | Impact                                         | Mitigation / present status                                                                                                                                                         |
|---------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lighting differs from calibration session         | Wrong colour / pass-side decision              | Verify the detector on the actual field in venue light before the run; the CIELab picker line retains a saved-calibration procedure as documented evidence of the same discipline.  |
| Pillar briefly blocks side LiDAR                  | False corner trigger                           | Round 2 corners trigger on the <b>front</b> LiDAR (20 cm) with a 1 s cooldown, so a pillar occluding a side sensor cannot fire a turn; Round 1 uses a rising edge + 1 s lidar mute. |
| Vision process stalls                             | Obstacle logic unavailable                     | ESP32 motion loop remains independent; robot continues on the last valid heading target.                                                                                            |
| Frame flex loosens wheels                         | Mechanical failure / no run                    | Resolved by rigid LEGO space-frame redesign; later sustained runs had no wheel loss.                                                                                                |
| Servo powered through logic rail                  | Voltage disturbance / reset risk               | Servo now uses dedicated 5 V rail with common ground; ESP32 provides signal only.                                                                                                   |
| I <sup>2</sup> C address collision among TF-Lunas | Sensors cannot be distinguished                | PCA9548A isolates each 0x10 TF-Luna on a separate channel.                                                                                                                          |
| Wireless active during a round (rule 11.10)       | Rule violation                                 | Procedure: disable Pi Wi-Fi/Bluetooth before every round; the competition stack opens no network paths — preview is a local window, headless by default.                            |
| Round 2 final integration incomplete              | Unknown competition reliability                | Final timed Round 2 test scheduled as the critical pre-competition validation task.                                                                                                 |
| End-of-run parking not validated                  | Lost parking points / possible limiter contact | Parking-lot exit S-curve is in the competition firmware; the park-in shimmy controller (branch feat/round2-park) is written but not yet run — validate conservatively or scope out. |

### Iteration pattern used by the team

- Identify the observed failure or constraint.
- Record the failure before changing the design.
- Form a specific hypothesis about the cause.
- Change one subsystem or procedure that addresses the cause.
- Repeat the relevant test and compare with the earlier evidence.
- Keep rejected approaches in the repository when they explain a trade-off or prevent the team from repeating the same mistake.

## 12 Criterion 5 — Reproducibility & GitHub Quality

Another team should be able to understand, rebuild and run the system from the repository

### Repository structure

| Path                  | Purpose                                                                                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| docs/                 | Five WRO criterion documents, predicted power budget, testing workflow, evaluation raw data, figures and photos.                                                                                                                                                                                                    |
| src/Round 1/          | ESP32 Open Challenge firmware: heading hold, rising-edge corner detection, steering and drive logic (+ ultrasonic prototype, legacy IMU test).                                                                                                                                                                      |
| src/Round 2/          | <b>competition/ — the flashed Nationals stack (test.cpp + detector.py + detect.py, with its own README);</b> plus the 3-class ONNX detector line (weights and .pt in detector/models/), Pi sender, earlier ESP32 obstacle controllers, evaluation harness, vision A/B tools, and preserved lineages under history/. |
| models/               | Team CAD: printed-part STLs (motor clamp, servo horn beam, TF-Luna enclosure), LEGO chassis capture (.lxfml) and build instructions.                                                                                                                                                                                |
| schemes/              | Signal and power wiring diagrams (v0.2, PNG + PDF, generated by a committed script) and the as-built circuit photograph.                                                                                                                                                                                            |
| t-photos/ · v-photos/ | Team photographs (official + informal); vehicle from all six views plus detail shots.                                                                                                                                                                                                                               |
| video/                | Required autonomous performance video links (one per challenge, ≥30 s each).                                                                                                                                                                                                                                        |
| other/                | Datasheets, setup notes and engineering test footage (test runs of 8–10 Aug).                                                                                                                                                                                                                                       |
| tools/                | Host-side utilities: Pi dataset capture rig (wro_capture.py) and serial logging.                                                                                                                                                                                                                                    |

### Software environment needed to reproduce the robot

| Part                  | Environment                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESP32 firmware        | Arduino IDE or PlatformIO; Adafruit BNO055, Adafruit Unified Sensor, Adafruit BusIO and ESP32Servo.                                                                               |
| Raspberry Pi vision   | Python 3.10–3.12; NumPy, OpenCV, <b>onnxruntime</b> , PyAV, pyserial, Pillow, SciPy and v4l2 utilities. Detector weights (.onnx + .pt) are versioned in-repo.                     |
| Sensor setup          | Three TF-Lunas must be configured to I <sup>2</sup> C mode before use; all remain at address 0x10 and are separated by the PCA9548A.                                              |
| Per-venue calibration | Verify detector per-class recall on the actual field; for the fallback picker, run the colour calibration routine in venue lighting, save, and reuse the saved values headlessly. |

### Competition artifacts

| Artifact                              | Required | Status                                                                    |
|---------------------------------------|----------|---------------------------------------------------------------------------|
| README > 5000 characters              | Yes      | Present (10.3k), with team roles, documentation index and version history |
| Source code for programmed components | Yes      | Present, src/                                                             |
| Vehicle photos from multiple views    | Yes      | Present — six views + detail shots                                        |
| Team photo                            | Yes      | Present — formal and informal                                             |
| Wiring information                    | Yes      | Present in schemes/; power overview in §05                                |

|                                           |     |                                                                                                           |
|-------------------------------------------|-----|-----------------------------------------------------------------------------------------------------------|
| CAD / model / chassis rebuild information | Yes | Present — STLs, .lxfml, build instructions; renders in §03                                                |
| Open Challenge video                      | Yes | <a href="https://youtu.be/PbOQJ54wr-o">https://youtu.be/PbOQJ54wr-o</a> (33 s, uploaded 12 Aug)           |
| Obstacle Challenge video                  | Yes | <a href="https://youtu.be/dBjqMQ-jRBA">https://youtu.be/dBjqMQ-jRBA</a> (100 s, uploaded 12 Aug)          |
| Commit history                            | —   | 100+ commits with conventional messages, PR-based merges, release tags (vo.2.0, vo.3.0) and release notes |

GitHub repository: <https://github.com/teddriveomo/wro-2026-future-engineers> (mirror kept byte-identical at [github.com/comede547/wro-2026-future-engineers](https://github.com/comede547/wro-2026-future-engineers)).

## 13 Testing & Final Pre-Competition Status

*What has been demonstrated, what remains pending, and what should be frozen before 26 August*

| Validation item                    | Status on 24 Aug 2026                         | Evidence / next step                                                                                                                                                                              |
|------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Final robot mass                   | DONE — 785 g                                  | Below 1.5 kg limit by 715 g.                                                                                                                                                                      |
| Open Challenge three laps          | DONE — 24 s                                   | Latest team-reported full three-lap result; cross-checked against the drivetrain model in §04.                                                                                                    |
| Wheel retention / chassis rigidity | DONE                                          | Post-redesign sustained runs recorded without wheel loss; final frame retained.                                                                                                                   |
| Main power switch                  | DONE — installed                              | Use together with one separate start button for the WRO start sequence.                                                                                                                           |
| Round 2 controller / perception    | COMMITTED — competition stack                 | src/Round 2/competition/ (test.cpp + detector.py); final timed end-to-end track test still pending. Pre-flight: the model file next to detector.py must match ONNX_MODEL_PATH ("best_ncnn.onnx"). |
| Final Round 2 timed test           | PENDING                                       | Run on competition field / equivalent mat and record success, time, obstacle-side errors and stop behaviour.                                                                                      |
| Parking                            | EXIT: IN FIRMWARE · PARK-IN: WRITTEN, NOT RUN | Boot S-curve exit is part of the competition firmware; park-in shimmy controller on feat/round2-park — validate conservatively before enabling.                                                   |
| Loaded straight-line speed         | RE-MEASURE                                    | Tape + timer over a marked distance before rounds; until then the journal carries only the derived 1.15 m/s unloaded ceiling (§03). The same run measures the driven lap line (§04).              |
| Axle-load split                    | MEASURE                                       | Front/rear weight split settles the traction-vs-stall regime (§03).                                                                                                                               |
| Radios off (rule 11.10)            | PROCEDURE                                     | Disable Pi Wi-Fi/Bluetooth before every round; verify before staging.                                                                                                                             |

### Recommended final test record for Round 2

- Run at least 5–10 full attempts with mixed red/green pillar arrangements.
- For every run record: completion yes/no, total time, wrong-side pillar events, moved pillars, wall contacts, number of recoveries/reverses and whether the robot stopped correctly after three laps.
- For parking record: bay detected yes/no, entry attempt yes/no, limiter touch yes/no, fully/partly parked and final orientation.
- Freeze the exact source revision used for the final successful run and commit the result table/video/log with a clear message.

**Competition dates.** 26–28 August 2026. The final Round 2 and parking tests are therefore the highest-priority remaining engineering tasks. Results should be added to the repository and this journal only after they are actually observed.

## 14 Evidence Map to the WRO 2026 Rubric

Where each judging criterion is supported in this journal and repository

| WRO criterion                             | Evidence provided                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobility & Mechanical Design              | Final chassis and photos; Ackermann steering; MG90 range/centre; N2O + TB6612 drivetrain; 20T:28T ratio; wheel dimensions; final mass;<br><b>torque/traction/acceleration derivation; CAD of printed parts; failure-driven frame redesign; Open three-lap test.</b>                                                       |
| Power & Sensor Architecture               | As-built power tree + circuit photo; installed main switch; separated motor/servo/electronics paths; predicted power budget; 3× TF-Luna + PCA9548A; BNO055; camera; placement/beam-width reasoning with POV imagery; calibration and noise controls.                                                                      |
| Software Architecture & Obstacle Strategy | Open flowchart; <b>verbatim firmware logic excerpt; P-vs-PID control design rationale with the documented sign-fix</b> ; 12-turn stop; Round 2 state machine; <b>perception pipeline flowchart with verbatim Python decision code</b> ; two-stack strategy with on-site selection criterion; explicit pending validation. |
| Systems Thinking & Engineering Decisions  | Constraints incl. rule 11.10; alternatives table; dataset leakage audit and withdrawal of numbers; neural→classical→retrained-neural evolution; chassis stiffness diagnosis; risk register; measured reasons for decisions.                                                                                               |
| Reproducibility & GitHub Quality          | Public repository + byte-identical mirror; role-attributed README; code incl. versioned weights; CAD/STLs; wiring; photos; required videos; environment; calibration setup; test workflow; 100+ commit history with PR merges and release tags.                                                                           |

### Final team checklist

- ☒ Engineering story explains why decisions were made, not only what was assembled.
- ☒ Mechanical design, steering, drivetrain and torque budget are illustrated and reproducible.
- ☒ Power distribution and sensor placement are documented with imagery.
- ☒ Software control flow and obstacle strategy are understandable without reading raw source first — and the key source is excerpted verbatim.
- ☒ Multiple iteration cycles (chassis, steering sign, dataset split, detector lineage) are supported by test evidence.
- ☒ GitHub includes code, wiring, models/build information, photos, videos and meaningful history.
- ☐ Final Round 2 timed validation must still be added after it is performed.
- ☐ Parking validation must still be added after it is performed.
- ☐ Loaded speed + driven-line tape measurements before the competition copy is frozen.

### Conclusion

The final TED Drive vehicle is the result of repeated engineering iteration rather than a single build. Mechanical flex changed the chassis, an inverted sign changed the steering law, data leakage changed the validation method, poor field transfer changed the vision architecture twice, and power/sensor integration changed the wiring. The final Open Challenge system has completed three laps in 24 seconds on the latest reported test — a figure this journal cross-checks against its own drivetrain model rather than asserting — and the complete vehicle weighs 785 g. The remaining work is clearly bounded: finish timed Round 2 validation, complete parking, and take the two single-session measurements (loaded speed, axle-load split) that close the mechanical chapter. Keeping these items explicit is part of the engineering process and makes the documentation reproducible and credible.

## References

- World Robot Olympiad Association, *WRO Future Engineers 2026 — General Rules*, version 15 January 2026, especially Appendix C: Engineering Journal evaluation and the Documentation Rubric.
- Team TED Drive public GitHub repository: <https://github.com/teddriveomo/wro-2026-future-engineers>
- Open Challenge video: <https://youtu.be/PbOQJ54wr-o> · Obstacle Challenge video: <https://youtu.be/dBjqMQ-jRBA>

**Journal status.** Final pre-competition edition rev 3, prepared 24 August 2026 from the public repository at commit b715a4b, the WRO 2026 Appendix C rubric and the team's latest supplied measurements/photos. Pending test items are intentionally labelled as pending rather than presented as completed results.